Bounding Treatment Effects with Symmetry Sensitivity Models*

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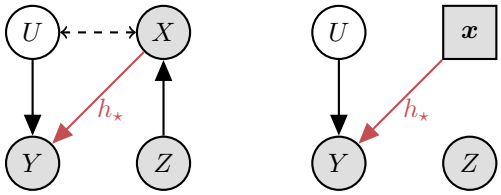
Code: https://github.com/uzairakbar/symmetry4CausalBounds
Project Page: https://uzairakbar.com/symmetry4CausalBounds

Abstract

Partial identification (PI) entails estimating bounds on causal effects by encoding different assumptions on data generation as a constrained optimization problem. Such bounds can suffice to inform policy decisions even if the causal effect itself is not identifiable. Often vacuous in practice, practitioners seek to exhaustively encode domain knowledge as additional constraints to make the PI bounds more informative. We introduce known *data symmetries*—invariance of the causal effect under certain data transformations—as a new source of constraints to inform PI. We operationalize this as a *shape constraint* on the causal function, and via a *change of measure* against which PI is posed using simple data pre-processing. Both approaches are shown to sharpen bounds under two canonical PI models. This is shown both theoretically for the population case, and via experiments in the finite-sample case. More broadly, our framework establishes data symmetries as a natural, underutilized source of background knowledge for robust causal inference.

1 PRELIMINARIES

Statistical vs. causal inference. Given n samples $\mathcal{D} := \{(\boldsymbol{x}_i, y_i, \boldsymbol{z}_i)\}_{i\in[n]}$ from the observational distribution p_{XYZ} of treatment $X \in \mathcal{X} \subseteq \mathbb{R}^{d_{\boldsymbol{x}}}$, outcome $Y \in \mathcal{Y} \subseteq \mathbb{R}$ and pre-treatment co-variates $Z \in \mathcal{Z} \subseteq \mathbb{R}^{d_{\boldsymbol{z}}}$, we aim to find the function $h_\star \in \mathcal{H} := \{h : \mathcal{X} \to \mathbb{R}\}$ defining the causal effect of X on Y under the data generating process (DGP):



(a) Unobserved confounding model (b) Intervention $\mathrm{do}(\boldsymbol{x})$

Figure 1: *(a)* Graph of Eq. (1) with IV Z and hidden U confounds (X, Y). *(b)* Intervention removes this confounding.

$$Y = h_\star(X) + U + \xi, \qquad \xi \perp\!\!\!\perp (U, X, Z). \tag{1}$$

The unobserved noise (ξ, U) is assumed mean-centered w.l.o.g., with the intercept absorbed into $h_\star$. Independence of noise $U \perp\!\!\!\perp X$ implies $\mathbb{E}[Y \,|\, \boldsymbol{x}] = h_\star(\boldsymbol{x})$. *Statistical inference* entails identifying precisely the Bayes optimal predictor $\mathbb{E}[Y \,|\, \boldsymbol{x}]$ from $\mathcal{D}$ via *empirical risk minimization* (ERM): minimize an empirical version of the *statistical risk*,

$$R_{\mathrm{erm}}(h) := \mathbb{E}\big[(Y - h(X))^2\big], \tag{2}$$

Then, for a sufficiently rich hypothesis class, the minimizer h_{erm} over $\mathcal{H}$ is an unbiased estimate of $h_\star$ Vapnik [1998].

However, $U \perp\!\!\!\perp X$ may not hold in general, in which case $\mathbb{E}[U \,|\, \boldsymbol{x}] \neq 0$ and the conditional $\mathbb{E}[Y \,|\, \boldsymbol{x}]$ now gives a biased estimate of $h_\star(\boldsymbol{x})$ [Pearl, 2009, Peters et al., 2017]. This dependence arises due to unobserved common causes U of X and Y, known as *confounders*. We say that X and Y are confounded and refer to the resulting bias as *confounding bias* [Pearl, 2009]. The observed pre-treatment co-variates Z may contain auxiliary information, e.g. proxies for such confounders, that can help to mitigate this bias. This encompasses *causal inference* which seeks to identify $h_\star$ under hidden confounding, or at the very least account for it by finding bounds on $h_\star$ should point-identification not be possible. Both approaches are outlined below.

*This is a non-archival summary of the working paper Akbar et al. [2026], for discussion at the CDM Workshop @ UAI 2026.

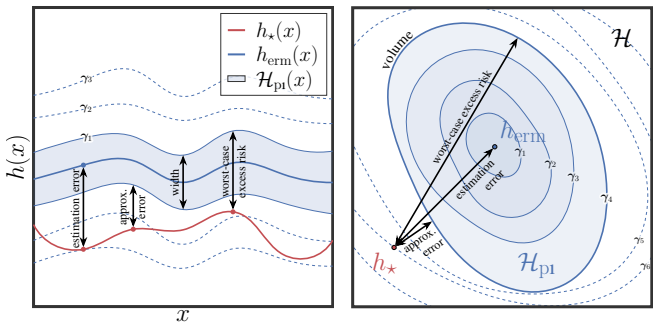


Figure 2: Depiction of PI sets $\mathcal{H}_{\text{PI}}(\boldsymbol{x})$ *(Left)* and $\mathcal{H}_{\text{PI}}$ *(Right)* for various sensitivity parameters $\boldsymbol{\gamma}$ and evaluation metrics.

Point-identification via intervention and adjustment. We can make treatment X and the confounder U independent via an *intervention* $\text{do}(X = \boldsymbol{x})$ that explicitly sets X to a fixed value $\boldsymbol{x}$ in Eq. (1) during data generation. Under such a treatment intervention, henceforth denoted as $\text{do}(\boldsymbol{x})$ for brevity, outcome Y follows the *interventional distribution* $p_{Y|\text{do}(\boldsymbol{x})}$. And the target estimand of Eq. (2) is now the *average treatment effect (ATE)* $\mathbb{E}[Y \,|\, \text{do}(\boldsymbol{x})]$, which equals $h_\star(\boldsymbol{x})$ for the DGP in Eq. (1). To capture the *estimation error* for a candidate hypothesis $h \in \mathcal{H}$, we use *causal excess risk* [Akbar et al., 2025, Vankadara et al., 2022]

$$E(h) \coloneqq \mathbb{E}\big[(h(X) - h_\star(X))^2\big].$$

Since interventions are often inaccessible for computing the ATE, so estimating $h_\star$ usually relies on access to the full joint distribution p of (U, X, Y, Z) via *back-door adjustment* $h^p_\star$ [Pearl, 2009, Peters et al., 2017, Xu and Gretton, 2022],

$$h^p_\star(\boldsymbol{x}) \coloneqq \mathbb{E}[\mathbb{E}[Y \,|\, \boldsymbol{x}, Z, U]], \qquad (U, X, Y, Z) \sim p.$$

Alternatively, $h_\star$ can be recovered via *instrumental variables (IVs)* Z satisfying (i) **relevance** $Z \not\perp\!\!\!\perp X$, (ii) **unconfoundedness:** $Z \perp\!\!\!\perp U$, (iii) **exclusion:** $Y \perp\!\!\!\perp Z \mid X, U$ [Zhang et al., 2023]. Hence, conditioning Eq. (1) on Z,

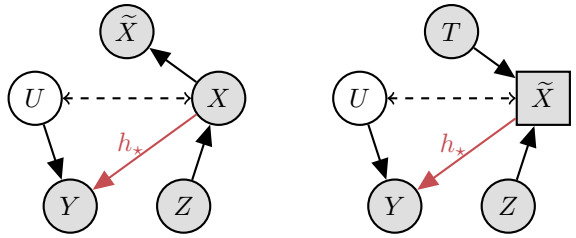
$$\mathbb{E}[Y - h_\star(X) \,|\, \boldsymbol{z}] = \mathbb{E}[U \,|\, \boldsymbol{z}] = 0. \tag{3}$$

An IV estimate h_{IV} solves Eq. (3) for $h_\star$, which now requires only the observational distribution p_{XYZ} [Sun et al., 2025].

Alas, Eq. (3) is typically under-specified, rendering $h_\star$ unidentifiable [Ailer et al., 2024]. Further, the IV properties themselves may be violated [Watson et al., 2024, Conley et al., 2012, Deaner, 2022, 2025]. And since U is unobserved, $h_\star$ is generally not recoverable from p_{XYZ} alone.

Partial identification (PI) offers a principled relaxation to this problem. Rather than insisting on assumptions sufficient for point identification, PI returns bounds on unidentifiable $h_\star$ which may suffice for decision making Padh et al. [2023].

Partial identification and sensitivity analysis. With the constraint set $\mathcal{P}_{\text{PI}}$ encoding our assumptions on the DGP Eq. (1), partial-identification considers all joints q s.t.



(a) Data augmentation on Eq. (1) (b) Soft intervention of Eq. (5)

Figure 3: For $\mathcal{T}$-inv. $h_\star$, *(a)* post-DA observational distribution equals *(b)* a soft intervention $\text{do}(X = \widetilde{X})$ in Eq. (1).

$$\mathcal{Q}_{\text{PI}}(p_{XYZ}) \coloneqq \Big\{q \in \mathcal{P}_{\text{PI}} : \; D_{\mathcal{F}}(p_V, q_V) = 0\Big\},$$
$$\text{where} \;\; D_{\mathcal{F}}(p_V, q_V) \coloneqq \sup_{f \in \mathcal{F}} \Big| \mathop{\mathbb{E}}_{p}[f(V)] - \mathop{\mathbb{E}}_{q}[f(V)] \Big|,$$

where some discrepancy metric $D_{\mathcal{F}}$, chosen here to be an *integral probability metric* (IPM) [Chau et al., 2025], matches q, p on observed variables $V \subseteq \{X, Y, Z\}$ over a set of witness functions $\mathcal{F} = \{f : \mathcal{V} \to \mathbb{R}\}$. If correctly specified, $p \in \mathcal{P}_{\text{PI}}$, and the following identified set $\mathcal{H}_{\text{PI}}$ contains the true function $h_\star$, and the interval $\mathcal{H}_{\text{PI}}(\boldsymbol{x})$ contains $h_\star(\boldsymbol{x})$,

$$\mathcal{H}_{\text{PI}} \coloneqq \big\{h^q_\star \colon q \in \mathcal{Q}_{\text{PI}}\big\}, \qquad \mathcal{H}_{\text{PI}}(\boldsymbol{x}) \coloneqq \big\{h^q_\star(\boldsymbol{x}) \colon q \in \mathcal{Q}_{\text{PI}}\big\},$$

where $\mathcal{Q}_{\text{PI}}$ abbreviates $\mathcal{Q}_{\text{PI}}(p_{XYZ})$. Computing interval $\mathcal{H}_{\text{PI}}(\boldsymbol{x})$ at $\boldsymbol{x}$ is often more practical, as for convex $\mathcal{Q}_{\text{PI}}$ it amounts to solving two constrained optimization problems

$$\mathcal{H}_{\text{PI}}(\boldsymbol{x}) = \big[\, \min_{q \in \mathcal{Q}_{\text{PI}}} h^q_\star(\boldsymbol{x}), \quad \max_{q \in \mathcal{Q}_{\text{PI}}} h^q_\star(\boldsymbol{x}) \,\big].$$

In either case, we want the identified sets to (i) contain the true solution, and (ii) be sharp, i.e. as small as possible. Of course these respectively depend on the validity and strength of assumptions that one is willing to make within $\mathcal{P}_{\text{PI}}$, with point-estimation now being a special case of singleton $\mathcal{Q}_{\text{PI}}$.

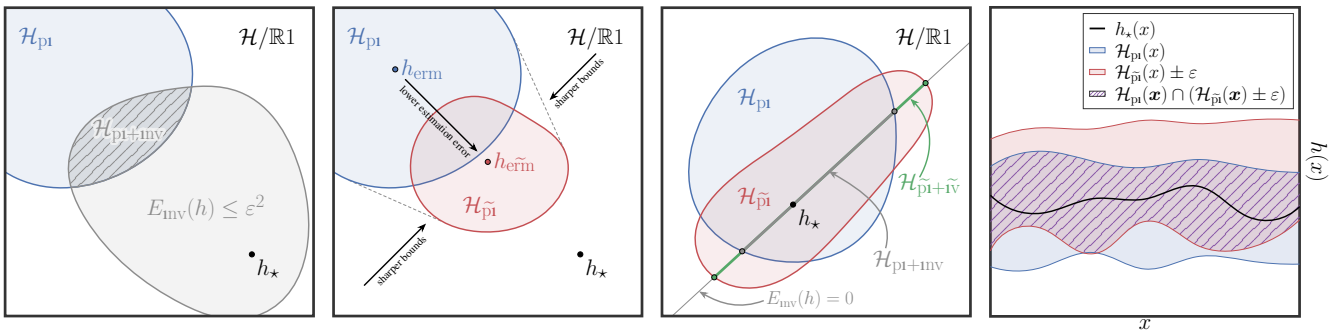
Consequently, one may also parameterize the constraint set as $\mathcal{P}_{\text{PI}}(\boldsymbol{\gamma})$ to conduct *sensitivity analysis* by varying parameters $\boldsymbol{\gamma}$ to see how $\mathcal{H}_{\text{PI}}(\boldsymbol{x})$ and $\mathcal{H}_{\text{PI}}$ evolve as assumptions are relaxed / tightened [Frauen et al., 2024, Penn et al., 2025].

While §B discusses existing choices for constructing $\mathcal{P}_{\text{PI}}$, the next section proposes our novel approach for the same.

2 KNOWN DATA SYMMETRIES

Symmetries are ubiquitous in scientific modeling, describing structural properties that do not change under certain data transformations Bronstein et al. [2021]. Formalized as a map $\alpha \colon \mathcal{X} \times \mathcal{T} \to \mathcal{X}$ parameterized by $\boldsymbol{\tau} \in \mathcal{T}$ with $\boldsymbol{\tau}\boldsymbol{x} \coloneqq \alpha(\boldsymbol{x}, \boldsymbol{\tau})$ for short, we say that $h_\star$ is $\mathcal{T}$*-invariant* if

$$h_\star(\boldsymbol{\tau}\boldsymbol{x}) = h_\star(\boldsymbol{x}), \qquad \forall\, (\boldsymbol{x}, \boldsymbol{\tau}) \in \mathcal{X} \times \mathcal{T}.$$



(a) $\mathcal{H}_{\text{PI+INV}}$. (b) $\mathcal{H}_{\widetilde{\text{PI}}}$. (c) $\mathcal{H}_{\widetilde{\text{PI+IV}}}$. (d) $\mathcal{H}_{\text{PI}}(\boldsymbol{x}) \cap (\mathcal{H}_{\widetilde{\text{PI}}}(\boldsymbol{x}) \pm \varepsilon)$

Figure 4: Illustration of identified sets: *(a)* Invariance shape constraint prunes the baseline set (§4). *(b)* DA pre-processing yields a sharper (Prop. 2), better-centered (Prop. 1) set (§5). *(c)* T-as-IV enforces symmetry (§D, cf. §E). *(d)* ε-pad-and-intersect dominates under arbitrary DA (Rem. 1).

This captures the notion that certain transformations preserve the causal relationship $X \to Y$. Examples include spatial translations in physics [Brandstetter et al., 2022], permutations of exchangeable units [Zaheer et al., 2017], and semantics-preserving text transformations [Ribeiro et al., 2020]. In image classification, for instance, a rotated image $\boldsymbol{\tau x}$ of a cat $\boldsymbol{x}$ still remains a cat [Cohen and Welling, 2016].

In practice, symmetries may hold only approximately, and $h_\star$ is ε-*approximately* $\mathcal{T}$-invariant if [Huang et al., 2023]

$$\sup_{\boldsymbol{x},\boldsymbol{\tau}} |h_\star(\boldsymbol{x}) - h_\star(\boldsymbol{\tau x})| \le \varepsilon.$$

Symmetry assumptions are well-established across many applications and routinely used to improve statistical estimation. For example, given some probability measure p_T on $\mathcal{T}$, *data augmentation* (DA) reduces estimation variance by generating multiple m augmentations $\{(\boldsymbol{\tau}_{i,j}\boldsymbol{x}_i,\ y_i)\}_{j\in[m]}$, with $\boldsymbol{\tau}_{i,j} \sim p_T$ per sample $(\boldsymbol{x}_i, y_i, \cdot) \in \mathcal{D}$, increasing data variability for statistical risk evaluation [Lyle et al., 2020, Chen et al., 2020, Savkin et al., 2020],

$$R_{\widetilde{\text{ERM}}}(h) \coloneqq \mathbb{E}\big[\ell\big(Y, h(\widetilde{X})\big)\big], \tag{4}$$

where $\widetilde{X} \coloneqq TX$ for $T \sim p_T$, as depicted in Fig. 3a.

Despite their success in statistical inference, symmetry priors remain understudied in causal inference [Akbar et al., 2025]. This gap is particularly consequential for PI, where bounds are often vacuous. Practitioners therefore seek to exhaustively impose constraints encoding domain knowledge to sharpen them. Symmetry assumptions hence provide a natural and powerful addition to this toolkit, especially in domains where they are already established. In what follows, we provide a first analysis of their application to sharpen PI.

3 SYMMETRY-INFORMED BOUNDS

4 A SYMMETRY SHAPE CONSTRAINT

Following the PI shape constraint tradition which restricts the class of causal function $h_\star$ Chetverikov et al. [2018], the most direct approach is to incorporate $\mathcal{T}$-invariance of $h_\star$ as a solver feasibility requirement:

$$E_{\text{INV}}(h) \coloneqq \mathbb{E}_p\big[|h(X) - h(\widetilde{X})|^2\big],$$
$$\mathcal{P}_{\text{PI+INV}} \coloneqq \big\{\ q \in \mathcal{P}_{\text{PI}} \ :\ E_{\text{INV}}(h_\star^q) \le \varepsilon^2\ \big\}.$$

Other *invariance-error* E_{INV} functionals may serve equally [Yang et al., 2019], as does restricting $\mathcal{H}$ to be invariant by construction [Cohen and Welling, 2016], though the latter can sit awkwardly with standard PI solvers.

$\mathcal{P}_{\text{PI+INV}}$ is compatible with modern deep learning-PI: $E_{\text{INV}}(\cdot)$ is differentiable and can be evaluated via Monte Carlo, so it plugs into gradient-based solvers as a regularizer [Padh et al., 2023, Kilbertus et al., 2020, Meresht et al., 2022, Hu et al., 2021]. There are however operational limitations: it requires white-box access to the solver's objective, precluding black-box or fixed-optimization software. Furthermore, optimizing non-convex invariance constraints for complex, high-dimensional data can be expensive and notoriously unstable [Schweisthal et al., 2025, Padh et al., 2023]. This motivates our second approach that touches only the data, and not the solver.

5 A CHANGE OF MEASURE UNDER SYMMETRY TRANSFORMATIONS

Where §4 constrains the PI candidates, we instead change the measure $p_X \to p_{\widetilde{X}}$ under which PI is posed while leaving the solver untouched, composing with it as follows

$$\mathcal{Q}_{\widetilde{\text{PI}}}(p_{XYZ}) \coloneqq \mathcal{Q}_{\text{PI}}(p_{\widetilde{X}YZ}).$$

This induces a reweighting of candidate evaluations, which may prune previously feasible functions. Crucially, symmetry now guarantees that validity continues to be preserved.

Intuitively, for a $\mathcal{T}$-inv. $h_\star$, the post-DA data $(\widetilde{X}, Y)$ follows

$$Y = h^\star(\widetilde{X}) + U + \xi, \tag{5}$$

thus simulating a *soft intervention* that maintains the causal relation $X \to Y$. Two things follow; (i) the model Eq. (5) has confounding no greater than the original Eq. (1), so bounded-confounding models remain valid. And (ii) the graph of the induced intervention Eq. (5) is identical to the original upto a relabeling of the nodes (Fig. 3a, cf. Prop. 4). Consequently, broad classes of graphical restrictions Zhang and Bareinboim [2021] and bounded confounding constraints Frauen et al. [2024] remain valid post-DA.

We take inspiration from IV-based PI methods, where "strong" instruments that induce greater exogenous variation in X yield sharper PI bounds than weak ones [Kilbertus et al., 2020, Padh et al., 2023]. DA introduces precisely such synthetic variation: by transforming X along directions that leave $h^\star$ unchanged, we inject variation independent of the confounding mechanism. This raises the natural question:

Can DA sharpen PI bounds the way IVs do?

We answer affirmatively. DA benefits PI in four key ways:

(i) **Statistical efficiency:** Most straightforwardly, DA grows effective sample size of data, mitigating sampling variation and finite-sample errors; key sources of uncertainty in PI [Imbens and Manski, 2004].

(ii) **Valid bounds (§C):** Crucially, $\mathcal{T}$-inv. of $h_\star$ guarantees bound validity post-DA whenever $\mathcal{P}_{\text{PI}}$ is valid.

(iii) **Sharper bounds (Prop. 2):** DA adds variation in X that is explicitly exogenous, and therefore cannot be attributed to the confounder U. This can reduce ambiguity in PI, which leads to sharper bounds.

(iv) **Informative bounds (Thm. 2):** An identified set is "informative" when it has low average error E over $\mathcal{H}_{\text{PI}}$. DA can improve this by re-centering the bounds around the truth and shrinking them towards it.

Even under misspecification, robustness is guaranteed: ε-padding the post-DA intervals $[a, b] \pm \varepsilon := [a - \varepsilon, b + \varepsilon]$ and intersecting with the baseline intervals $\mathcal{H}_{\text{PI}}(\boldsymbol{x}) \cap (\mathcal{H}_{\widetilde{\text{PI}}} \pm \varepsilon)$ prevents degradation while preserving gains (Thm. 3.A).

To make these results concrete, we instantiate PI under the partial r^2 sensitivity model of Cinelli and Hazlett [2020] (Asm. 2), which bounds the observed variation in (X, Y) that can be explained by the unobserved confounder U. Under standard conditions, this yields an ellipsoidal identified set centered at the ERM with radius governed by an effective confounding budget. DA changes the geometry of this ellipsoid; inflating some directions, deflating others; but the intersection with the baseline guarantees no regression.

Finally, the symmetry parameters T satisfy IV properties by construction Akbar et al. [2025]: un-confoundedness

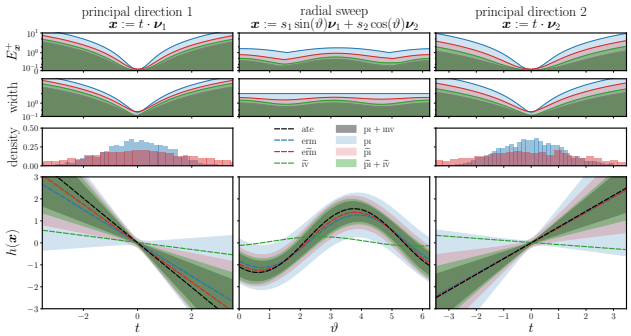


Figure 5: **Simulation data.** Across queries $\boldsymbol{x}$ along principal components $\boldsymbol{\nu}_1, \boldsymbol{\nu}_2$ of $\text{Var}(X)$, interval width *(Top)* decrease over baseline while maintaining validity *(Bottom)*.

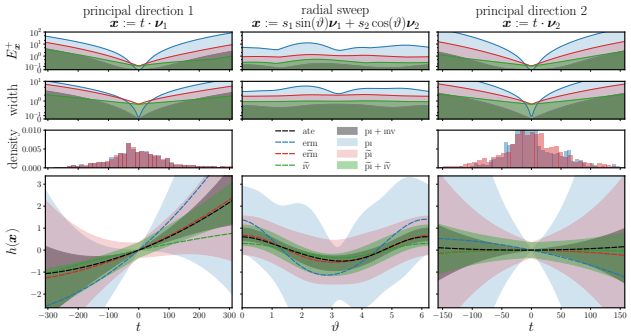


Figure 6: **Optical device data.** Real-world sharpening of PI via known symmetries; image orientation does not affect irradiance, rendering photo-diode measurements invariant.

$(T \perp\!\!\!\perp U)$ and exclusion $(Y \perp\!\!\!\perp T \mid \widetilde{X}, U)$. This allows enforcing additional invariance constraints within black-box IV-based PI solvers by treating T as an instrument (see §D).

6 EXPERIMENTS

We validate our methods in finite samples. To isolate symmetry-driven sharpening from standard data-augmentation gains, we keep the augmented dataset size fixed to the original sample size. Full experimental details are provided in §G.

Linear simulation. In the simulation, we generate confounded linear outcomes and construct transformations T that inject Gaussian noise in X only within the null-space of the true causal effect, so $h_\star$ is invariant but remains unidentified without observing the confounder. Figure 5 shows that symmetry-aware PI produces tighter intervals than the baseline across principal-direction and radial query sweeps, while preserving coverage of the target.

Optical device. For the optical-device experiment, we use the benchmark of Janzing and Schölkopf [2018], where hidden LED intensity confounds webcam-captured 3×3 images

and photo-diode voltage readings. We use image symmetries such as vertical/horizontal flips and 90° rotations as DA transformations. Figure 6 shows the same qualitative trend on real data: incorporating known symmetries yields sharper and more informative PI bounds.

7 DISCUSSION

Limitations. Our proposed framework provides a practical hierarchy of symmetry constraints for PI: should an invariance shape constraint pose implementation challenges, practitioners can resort to the DA operationalization. If they suspect misspecification by some ε, they need only to ε-pad bounds. Which leads us to the main limitation of our work: while the soft intervention framing in Eq. (5) makes composability of DA with a broad class of disparate PI solvers plausible under well-specified symmetries, it is yet unclear whether this should also be the case under ε-misspecification. Having addressed this question under two canonical, arguably the most popular PI classes (Asm. 2 bounded confounding and Asm. 3 leaky IVs), further investigation of additional models is left as future work.

Broader applicability. Our framework is compatible with *all* PI models via the shape constraint operationalization. The DA approach also naturally extends to any PI or sensitivity model that imposes bounded confounding constraints equivalent to Asm. 2 (MSM Tan [2006], RO Guo et al. [2022], Wasserstein Meresht et al. [2022], total-variation Padh et al. [2023], etc.). Under well-specified symmetries, DA also composes with other shape-constraints and auxiliary variable constraints other than IVs. For example, Asm. 2 can admit conditioning on pre-treatment Z Cinelli and Hazlett [2020], which we omit only for clarity of exposition. With invariance in the response function $H_\star(\boldsymbol{x}) = H_\star(\boldsymbol{\tau x})$, the soft-intervention framing in Eq. (5) also extends DA to response function models Padh et al. [2023], as well as IVs with latent U Kilbertus et al. [2020], proxy variables Ghassami et al. [2026], Mastouri et al. [2021], Sun et al. [2025] and others Meresht et al. [2022], Zhang and Bareinboim [2021].

Future work. We aim to generalize our change-of-measure framework to (i) general causal queries on $p_{Y\,|\,\mathrm{do}(\boldsymbol{x})}$ Ailer et al. [2024], (ii) equi-variance $H_\star(\boldsymbol{\tau x}) = \beta(\boldsymbol{x}, \boldsymbol{\tau}) H_\star(\boldsymbol{x})$ under cocycle β Dance and Bloem-Reddy [2025], and (iii) latent U under a c-Wasserstein notion of misspecification $\sup_{\boldsymbol{x},\boldsymbol{\tau}} W_c(p_{H_\star(\tau \boldsymbol{x})}, \beta(\boldsymbol{x}, \boldsymbol{\tau})_{\#} p_{H_\star(\boldsymbol{x})}) \leq \varepsilon$ Chen et al. [2020]. Also, it may be interesting to demonstrate application across causal settings like genomics [Weinstein and Blei, 2026] and medicine [Müller et al., 2021, Elaldi et al., 2023, Slote et al., 2025, Cheema, 2026].

Conclusion. Researchers are advised to incorporate established domain symmetries as a standard component of their PI toolkit. Integration is effortless with DA pre-processing.

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Bounding Treatment Effects with Symmetry Sensitivity Models (Supplementary Material)

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A LIST OF SYMBOLS

Symbol	Description
Variables and Spaces	
$X, \mathcal{X}, \boldsymbol{x}$	Treatment, its support, and a realization
$Y, \mathcal{Y}, y$	Outcome, its support, and a realization
$Z, \mathcal{Z}, \boldsymbol{z}$	Pre-treatment co-variates / instruments (IVs), their support, and realization
U, ξ	Unobserved confounder, exogenous noise (mean-zero, $\xi \perp\!\!\!\perp (U, X, Z)$)
$\widetilde{X}, \widetilde{Z}$	Post-DA treatment $\widetilde{X} \coloneqq TX$ and instrument $\widetilde{Z} \coloneqq (Z, T)$
Hypotheses and Estimators	
$h, h_\star, \mathcal{H}$	Candidate hypothesis, causal target, hypothesis class
$\mathcal{H}_X, \mathcal{H}_{\widetilde{X}}, k$	X-, $\widetilde{X}$-centered $\mathcal{H}$ with covariance forms $\langle\cdot,\cdot\rangle_X, \langle\cdot,\cdot\rangle_{\widetilde{X}}$ and $k \coloneqq \dim(\mathcal{H}_X)$
$h_{\mathrm{erm}}, h_{\widetilde{\mathrm{erm}}}, b, \tilde{b}$	ERM estimators $h_\bullet \coloneqq \operatorname{argmin}_{h\in\mathcal{H}} R_\bullet(h)$, their respective biases $h_\bullet - h_\star$
$h_{\mathrm{iv}}, h^p_\star$	IV estimator, and back-door adjustment over joint p
$g_{\boldsymbol{x}}, \tilde{g}_{\boldsymbol{x}}$	Riesz representers for point-evaluation $h \mapsto h(\boldsymbol{x})$ in $\mathcal{H}_X$ and $\mathcal{H}_{\widetilde{X}}$
$\mathcal{H}_{\mathrm{inv}}, \mathcal{I}, \mathcal{P}$	$\mathcal{P}$ orthogonally projects onto ambient invariant space $\mathcal{I}$, and $\mathcal{H}_{\mathrm{inv}} \coloneqq \mathcal{H} \cap \mathcal{I}$
Symmetries, Interventions and Change of Measure	
$T, \mathcal{T}, \boldsymbol{\tau}, \alpha$	Random transformation, support, realization, and defining map $\boldsymbol{\tau}\boldsymbol{x} = \alpha(\boldsymbol{x}, \boldsymbol{\tau})$
$\mathrm{do}(\boldsymbol{x}), \mathrm{do}(X = \widetilde{X})$	Hard intervention $\mathrm{do}(X = \boldsymbol{x})$, and DA-simulated soft intervention
ε, η	Invariance residual $\eta \coloneqq h_\star(X) - h_\star(\widetilde{X})$ with uniform budget $\lvert\eta\rvert \le \varepsilon$ a.s.
$\mathcal{S}, \mathcal{K}, \delta^*\delta$	Shift operator on $\mathcal{H}_{\widetilde{X}}$, splits into shape/variance + rank-1 level/mean shifts
$\sigma^2, \tilde{\sigma}^2, \rho$	Baseline / post-DA MMSE, and the noise-inflation factor $\rho \coloneqq \tilde{\sigma}^2/\sigma^2 \ge 1$
Constraints and Partial Identification	
p, q	True observational joint, and a candidate joint distribution in $\mathcal{Q}_{\mathrm{PI}}$
$D_{\mathcal{F}}, \mathcal{F}$	IPM and its witness-function class defining $\mathcal{Q}_{\mathrm{PI}}$
$\gamma_{\boldsymbol{x}}, \gamma_y, \gamma$	Confounding budgets for X, Y; effective budget $\gamma \coloneqq \gamma_{\boldsymbol{x}}\gamma_y/(1-\gamma_{\boldsymbol{x}})$
$\gamma_{\boldsymbol{z}}, \gamma^\star_{\boldsymbol{z}}, \tilde{\gamma}_{\boldsymbol{z}}$	Assumed, true, and adjusted IV violation budgets
$\mathcal{P}_{\mathrm{PI}}, \mathcal{P}_{\mathrm{PI+INV}}, \mathcal{P}_{\mathrm{PI+IV}}$	PI constraint sets: baseline, augmented with invariance / IV constraints
$\mathcal{Q}_{\mathrm{PI}}, \mathcal{Q}_{\widetilde{\mathrm{PI}}}, \mathcal{Q}_{\widetilde{\mathrm{PI}}+\widetilde{\mathrm{IV}}}$	Candidate joints: baseline, post-DA, and DA-as-IV
$\mathcal{H}_{\mathrm{PI}}, \mathcal{H}_{\widetilde{\mathrm{PI}}}, \mathcal{H}_{\mathrm{PI}}(\boldsymbol{x}), \mathcal{H}_{\widetilde{\mathrm{PI}}}(\boldsymbol{x})$	Baseline and post-DA identified sets, and their respective intervals at query $\boldsymbol{x}$
Risks and Evaluation Metrics	
$\ell(y, y')$	Proper, convex loss function (e.g., squared error)
$R_{\mathrm{erm}}(h), R_{\widetilde{\mathrm{erm}}}(h)$	Observational risks, pre- and post-DA
$R_{\boldsymbol{x}}(y), R(h)$	Causal / interventional risks, point-wise and averaged
$E_{\boldsymbol{x}}(y), E(h)$	Causal excess risk/ estimation error $E(h) = R(h) - R(h_\star)$, pointwise via $R_{\boldsymbol{x}}$
$E^-(\mathcal{H}), \overline{E}(\mathcal{H}), E^+(\mathcal{H})$	Approximation (min), averaged, worst-case (max) errors E over set $\mathcal{H}$
$E^-_{\boldsymbol{x}}(\mathcal{Y}), \overline{E}_{\boldsymbol{x}}(\mathcal{Y}), E^+_{\boldsymbol{x}}(\mathcal{Y})$	Point-wise versions of the above metrics via $E_{\boldsymbol{x}}$ over interval $\mathcal{Y}$
$E_{\mathrm{inv}}(h)$	Invariance error $\mathbb{E}_p\left[\lvert h(X) - h(\widetilde{X})\rvert^2\right]$

B CHOICE OF CONSTRAINTS

The nature and construction of $\mathcal{P}_{\text{PI}}$ depends on domain knowledge. At the most agnostic end, Manski's "no-assumption" bounds use only a bounded Y support to constrain $\mathcal{H}_{\text{PI}}(\boldsymbol{x})$ for a binary $\boldsymbol{x} \in \{0, 1\}$ [Manski, 1990]. Other popular approaches involve bounding the spurious correlation between X, Y, including the sensitivity model by Rosenbaum [2002] which parameterizes the strength of unmeasured confounding through odds ratios, its generalization of the Marginal Sensitivity Model (MSM) by Tan [2006] that does the same using propensity scores and the partial r^2 approach by Cinelli and Hazlett [2020] bounds the proportion of observed variance explained by unobserved confounders. More recently, [Bertsimas et al., 2023, Guo et al., 2022] formulated the PI problem as *robust optimization (RO)* over $\mathcal{Q}_{\text{PI}}$ constructed as a total variation ball around the observational distribution p_{XYZ}, and [Meresht et al., 2022] similarly uses Wasserstein constraints. An equivalent approach to modeling the confounding is to instead model the random function $H_\star(\cdot) \coloneqq h_\star(\cdot) + U + \xi$ itself, also known as the *response function* [Padh et al., 2023]. For example, [Hu et al., 2021] modeled these using generative adversarial networks to then match p_{XYZ} in distribution. Many of these methods also leverage auxiliary variables in addition to X, Y for imposing conditional independences to sharpen bounds [Ailer et al., 2024, Zheng et al., 2025]. Of note is IV based PI by Balke and Pearl [1997] where U may arbitrarily influence Y instead of additively like Eq. (1). Modern neural-network based variants for continuous, high-dimensional treatments and/or IVs are explored by [Kilbertus et al., 2020, Hu et al., 2021, Padh et al., 2023, Meresht et al., 2022, Gunsilius, 2020].

Perhaps more related to this work are *shape constraints* [Chetverikov et al., 2018]. These aim to sharpen PI by restricting the class of structural or response functions $h_\star, H_\star$. For example, by imposing positivity ($h_\star(\boldsymbol{x}) \geq 0$) [Leamer, 1981], monotonicity ($\boldsymbol{x}_1 \leq \boldsymbol{x}_2 \Rightarrow h_\star(\boldsymbol{x}_1) \leq h_\star(\boldsymbol{x}_2)$) [Manski, 1997, Manski and Pepper, 2000], convexity [Freyberger and Horowitz, 2015], curvature [Melnychuk et al., 2023], smoothness [Kim et al., 2018], sparsity [Pfister and Peters, 2022], or other regularities [Meresht et al., 2022], each narrowing the set of compatible DGPs to a convex feasible set. In the same vein, this work proposes known structural invariances of $h_\star$ as a natural shape constraint.

C A CHANGE OF MEASURE UNDER SYMMETRY TRANSFORMATIONS

DA as a change of measure. Where §4 constrains the feasible set, we instead change the measure $p_{XYZ} \to p_{\widetilde{X}YZ}$ under which PI is posed while leaving the solver untouched, composing with it as

$$\mathcal{Q}_{\widetilde{\text{pi}}}(p_{XYZ}) \coloneqq \mathcal{Q}_{\text{pi}}(p_{\widetilde{X}YZ}).$$

This change of measure reweights candidate evaluations, pruning previously feasible functions while, crucially, preserving validity across broad families of standard solvers. We now isolate the two mechanisms driving these effects: variance inflation (sharpening) and simulated intervention (validity).

Fit vs. evaluation geometries. Center Y w.l.o.g., then $\mathbb{E}[h_\star(X)] = 0$. So estimation of $h_\star$ only lives in the centered *shape space* $\mathcal{H}_X \coloneqq \{h \in \mathcal{H} \colon \mathbb{E}[h(X)] = 0\}$, equipped with covariance forms

$$\langle f, g\rangle_X \coloneqq \mathrm{Cov}(f(X), g(X)), \qquad \|h\|_X^2 \coloneqq \mathrm{Var}(h(X)),$$

and $(\mathcal{H}_{\widetilde{X}}, \langle\cdot,\cdot\rangle_{\widetilde{X}})$ likewise under $p_{\widetilde{X}}$. Of course, all errors are still reported in the ambient $L_2(X)$-norm; in particular, the causal excess risk under a sq.-loss $\ell(a,b) \coloneqq (a-b)^2$ decomposes orthogonally,

$$E(h) = \left\|h - h_\star\right\|_{L_2(X)}^2 = \underbrace{\|h - h_\star\|_X^2}_{\text{shape; lives on } \mathcal{H}_X} + \underbrace{\left(\mathbb{E}[h(X)] - \mathbb{E}[h_\star(X)]\right)^2}_{\text{level; lives off } \mathcal{H}_X}.$$

Fitting in $(\mathcal{H}_{\widetilde{X}}, \langle\cdot,\cdot\rangle_{\widetilde{X}})$-geometry can hence change the shape term. But $\widetilde{X}$-centered candidates $\mathcal{H}_{\widetilde{X}}$ need not be X-centered, so a level penalty appears. This split forces a separate treatment of covariance and mean. To make this analysis tractable, we first invoke the following regularity conditions on $\mathcal{H}$:

Assumption 1. Let $\mathcal{H} \subseteq L_2(X) \cap L_2(\widetilde{X})$ be closed under constant shifts $\mathcal{H} + \mathbb{R}1 = \mathcal{H}$. Optionally:
(a) Realizability: Suppose $\mathcal{H}$ contains both $h_\star(X)$, $\mathbb{E}[Y|X]$, and post-DA $\mathbb{E}[h_\star(X)|\widetilde{X}]$, $\mathbb{E}[Y|\widetilde{X}]$.
(b) Absolute continuity: $p_X \ll p_{\widetilde{X}}$, i.e. the post-DA distribution covers the baseline distribution.
(c) Finite dim. Hilbert: $\mathcal{H}$ is linear with $\dim(\mathcal{H}_X) = k < \infty$, and $\langle\cdot,\cdot\rangle_X$ positive-definite over $\mathcal{H}_X$.

Any class with a free intercept satisfies the base clause. Clause (a) removes statistical approx. bias to isolate confounding bias, our focus. Clause (b) is essentially free under DA; sample $\mathrm{Id} \in \mathcal{T}$, or mix-in p_{XYZ}. It lets the $\mathcal{H}_X$ Hilbert structure carry over to $\mathcal{H}_{\widetilde{X}}$. Clause (c) gives bounded evaluation $h(\boldsymbol{x})$. Without comparable regularity, PI intervals $\mathcal{H}_{\text{pi}}(\boldsymbol{x})$ for continuous X are known to be vacuous Kilbertus et al. [2020], Pearl [1995], Bonet [2001], Gunsilius [2020]. Crucially, finite dimension is only needed to compute set volume/avg.-error in closed form.

Dissecting the change of measure. Under Asms. 1b–1c, a unique, positive, self-adjoint $\mathcal{S} : \mathcal{H}_{\widetilde{X}} \to \mathcal{H}_{\widetilde{X}}$ pulls the evaluation $L_2(X)$-inner product back into the solver's native $(\mathcal{H}_{\widetilde{X}}, \langle\cdot,\cdot\rangle_{\widetilde{X}})$-geometry,

$$\langle f, \mathcal{S}g\rangle_{\widetilde{X}} = \langle f, g\rangle_{L_2(X)}, \quad \forall f, g \in \mathcal{H}_{\widetilde{X}}.$$

The shift operator $\mathcal{S}$ captures the fit vs. evaluation distribution shift. Its Rayleigh quotient $\mathcal{S}[h] \coloneqq \langle h, \mathcal{S}h\rangle_{\widetilde{X}} / \|h\|_{\widetilde{X}}^2$ ($\mathcal{S}[0] \coloneqq 0$) captures the same along direction h Horn and Johnson [1985]. Following $L_2(X)$-norm, its pull-back $\mathcal{S}$ also admits a shape/level split into operators $\mathcal{K} \colon \mathcal{H}_{\widetilde{X}} \to \mathcal{H}_{\widetilde{X}}$ and $\delta \colon \mathcal{H}_{\widetilde{X}} \to \mathbb{R}$ defined as

$$\langle f, \mathcal{K}g\rangle_{\widetilde{X}} = \langle f, g\rangle_X, \qquad \delta h = \mathbb{E}[h(X)] - \mathbb{E}[h(\widetilde{X})], \qquad \forall f, g, h \in \mathcal{H}_{\widetilde{X}}.$$

Lemma 1 (shift decomposition)**.** *Under Asms. 1b–1c,* $\mathcal{S} = \mathcal{K} + \delta^*\delta$*, with* $\mathcal{K} \succ 0$*,* $\delta^*\delta \succcurlyeq 0$ *s.t.* $\mathrm{rank}(\delta^*\delta) \leq 1$*, and* $\mathrm{tr}(\delta^*\delta) = D_{\mathcal{B}}^2(p_X, p_{\widetilde{X}})$ *is IPM² over the unit ball* $\mathcal{B} \coloneqq \{h \in \mathcal{H}_{\widetilde{X}} : \|h\|_{\widetilde{X}} \leq 1\}$.

The *variance-shift operator* $\mathcal{K}$ governs how the estimation geometry changes between $(\mathcal{H}_X, \langle\cdot,\cdot\rangle_X)$ and $(\mathcal{H}_{\widetilde{X}}, \langle\cdot,\cdot\rangle_{\widetilde{X}})$. By inflating data variance ($\mathcal{K} \preccurlyeq \mathrm{Id}$), DA contracts the evaluation geometry ($\mathcal{S} \preccurlyeq \mathrm{Id}$) and can reduce estimation error. However, the induced *out-of-distribution* (OOD) shift also incurs a penalty through the rank-1 *mean-shift operator* $\delta^*\delta \succcurlyeq 0$. By Lem. 1, its trace equals the sq.-IPM over a unit $\widetilde{X}$-ball, quantifying the cost of OOD shift. Thus, improvement requires variance inflation to outweigh the mean-shift ($\mathcal{K} \preccurlyeq \mathrm{Id} - \delta^*\delta$); a trade-off made concrete in subsequent results.

Beyond geometry, an informational quantity also governs PI; the observable noise level, captured by the minimum mean squared errors (MMSE) $\sigma^2 \coloneqq \min_{h\in\mathcal{H}} R_{\text{erm}}(h)$ and $\tilde{\sigma}^2 \coloneqq \min_{h\in\mathcal{H}} R_{\widetilde{\text{erm}}}(h)$. Often used to calibrate unobserved variables

in PI Deaner [2022], Cinelli and Hazlett [2025, 2020], they determine the intrinsic scale of the identified sets. Under Asm. 1a, the *data processing inequality* (DPI) on the chain $Y \leftrightarrow X \to \widetilde{X}$ gives $\rho := \tilde{\sigma}^2 / \sigma^2 \geq 1$, i.e. noisy $\widetilde{X}$ cannot predict Y better than X. We call ρ the *noise-inflation factor*.

We now explore how each of the geometric shifts $\mathcal{K}, \delta$, and the DPI $\rho \geq 1$ enters PI and sharpens it.

Fragility of point-estimation. To motivate our PI application of DA, we start by understanding its limitations under point-estimation. Notice that under $\mathcal{T}$-inv. $h_\star$, the post-DA data $(\widetilde{X}, Y)$ follows the model in Eq. (5). DA thus simulates a *soft* intervention $\mathrm{do}(X = \widetilde{X})$ on Eq. (1) Akbar et al. [2025], Massidda et al. [2023], Jaber et al. [2020], Besserve and Schölkopf [2022]; the link $U \leftrightarrow X$ is not severed but diluted by the DPI $\mathrm{Var}(\mathbb{E}[U|X]) \geq \mathrm{Var}(\mathbb{E}[U|\widetilde{X}])$ on the chain $U \leftrightarrow X \to \widetilde{X}$. For $\mathcal{T}$-inv. $h_\star$, this is exactly $\rho \geq 1$ (Prop. 4), which led Akbar et al. [2025] to propose DA for point-estimation:

Proposition 1 (estimation with DA). *For $\mathcal{T}$-inv. $h_\star$, Asms. 1a–1c, with post-DA bias $\tilde{b} := h_{\widetilde{\mathrm{erm}}} - h_\star$,*

$$E(h_{\mathrm{erm}}) - E(h_{\widetilde{\mathrm{erm}}}) \quad = \quad \underbrace{\sigma^2 \mathcal{S}[\tilde{b}] \cdot (\rho - 1)}_{\textit{gain}} + E(h_{\mathrm{erm}}) \cdot \Big(\underbrace{1 - \mathcal{K}[\tilde{b}]}_{\textit{lever}} - \underbrace{(\delta^* \delta)[\tilde{b}]}_{\textit{penalty}} \Big).$$

Proposition 1 reveals three fragilities: (i) **bias-direction dependence:** the DPI $\rho \geq 1$ alone does not guarantee improvement, which depends on whether $\mathcal{S}[\tilde{b}] \leq 1$ along the unobserved bias direction $\tilde{b}$, (ii) **misspecification:** if symmetry is misspecified, the gain term may itself become a loss, and (iii) **identifiability:** $h_\star$ remains unidentifiable regardless. All three defects are manageable under PI.

Partial identification with DA. In IV methods, "strong" IVs that induce more exogenous variation in X yield sharper PI bounds than weak ones [Kilbertus et al., 2020, Padh et al., 2023]. This motivates the question driving this section:

Can the synthetic exogenous variation introduced by DA likewise sharpen PI?

We answer yes, DA yields three PI guarantees, each resolving one fragility above (Table 2). **Validity:** The soft intervention Eq. (5) preserves the structural equation (1) and its causal graph (Fig. 3, cf. Prop. 4) up to relabeling $X \to \widetilde{X}$, while diluting confounding. Hence, any shape constraint Chetverikov et al. [2018], graphical restriction Zhang and Bareinboim [2021] or confounding bound Frauen et al. [2024] valid under p_{XYZ} remains valid under $p_{\widetilde{X}YZ}$. Crucially, validity follows from $\mathcal{T}$-invariance alone (Thm. 1), addressing (iii). **Sharpness / informativeness:** PI uncertainty is an *attribution problem*; the solver must split observed X, Y covariation between $h_\star$ and U. Only variation in X, Y whose relation to U is *known*, e.g. an IV Z, can break the tie. DA adds exactly such variation since T is exogenous by construction and cannot be attributed to U. This reduces ambiguity in PI, giving sharper, more informative bounds. Notably, this sharpening is governed by global quantities ($\rho\,\mathrm{tr}(\mathcal{S})/k$; Prop. 2 and Thm. 2), making sufficient conditions observable, unlike (i). **Robustness.** To make these guarantees robust to ε-misspecified symmetry, we deploy a simple post-processing: ε-pad intervals and intersect with baseline (Thm. 3), resolving (ii).

Beyond these population effects, DA has the familiar statistical-efficiency benefit of growing effective sample size, mitigating sampling variation and finite-sample error, key sources of PI uncertainty [Imbens and Manski, 2004]. To make these claims explicit, we instantiate PI under the bounded-confounding sensitivity model:

Assumption 2 (a bounded confounding sensitivity model Cinelli and Hazlett [2020]). For *confounding budgets* $\gamma_{\boldsymbol{x}}, \gamma_y$, let

$$\mathcal{P}_{\mathrm{PI}} := \left\{ q : \begin{array}{c} r^2_{U \sim X} \leq \gamma_{\boldsymbol{x}}, \\ r^2_{U \sim Y|X} \leq \gamma_y \end{array} \right\}, \qquad r^2_{A \sim B|C} := 1 - \frac{\mathbb{E}[\mathrm{Var}(A \,|\, B, C)]}{\mathbb{E}[\mathrm{Var}(A \,|\, C)]} \leq 1,$$

where $r^2_{A \sim B|C}$ is the *partial-r^2*, i.e. fraction of variation in A explained by B after controlling for C. Further, make the IPM $D_{\mathcal{F}}$ test all quadratic functions $\mathcal{F} := \{ f^2 : f \in \mathrm{span}(\mathcal{H}) + \mathbb{R}Y \}$ of (X, Y).

Assumption 2 adopts the widely used partial-r^2 model Cinelli and Hazlett [2020], a conceptual generalization of the classic Rosenbaum [2002]. While we employ this model in our analysis, we do not necessarily restrict ourselves to it: several PI families in §B impose equivalent constraints, reducing to ellipsoidal sets of the form

Lemma 2 (characterizing the identified set). *Under Asm. 2 and 1a or 1c, $\mathcal{H}_{\mathrm{PI}}$ is the covariance ball*

$$\mathcal{H}_{\mathrm{PI}} = \Big\{ h \in \mathcal{H}_X \,:\, \|h - h_{\mathrm{erm}}\|^2_X \leq \sigma^2 \gamma \Big\},$$

within $\mathcal{H}_X$*, centered at* h_{erm} *with radius* $\sigma\sqrt{\gamma}$*, and an* effective confounding budget *of* $\gamma := \frac{\gamma_{\text{x}}\gamma_y}{1-\gamma_{\text{x}}}$.

The post-DA analysis now reduces to three questions about this ellipsoid: whether it covers $h_\star$, does its size contract, and if its center/radius tradeoff lowers avg./worst-case errors. We treat these in turn.

Table 2: Point vs. *partial* identification under DA.

DA property	Identification metric	Safeguard
$\mathcal{T}$-inv. $h_\star$ $\Longrightarrow$	*validity* (Thm. 1) ✓	ε-pad (Thm. 3)
$\mathcal{S}[\tilde{b}] \leq 1$ $\Longrightarrow$	estimation error ↓ (Prop. 1)	✗
$\rho\,\mathrm{tr}(\mathcal{S})/k \leq 1 \Longrightarrow$	*informativeness* ↑ (Thm. 2) ⇑ *sharpness* ↑ (Prop. 2)	intersect with baseline set (Rem. 1)

C.1 VALID BOUNDS WITH DATA AUGMENTATION

First and foremost, we establish validity of post-DA bounds. In terms of approx. error, we show:

Theorem 1 (valid bounds with DA)**.** *Let* $h_\star$ *be* $\mathcal{T}$*-inv. under Asm. 1a and 2. Then, the post-DA set* $\mathcal{H}_{\widetilde{\text{pi}}}$ *is exactly valid whenever the baseline set* $\mathcal{H}_{\text{pi}}$ *is* ϵ*-valid with* $\epsilon \leq \sigma^2\big(\sqrt{\rho - 1 + \gamma\rho} - \sqrt{\gamma}\big)^2$.

Consequently, $\mathcal{H}_{\widetilde{\text{pi}}}$ *repairs* the ϵ-validity of $\mathcal{H}_{\text{pi}}$ since the fixed budget γ can now accommodate the diluted bias. Crucially, this follows from the simulated intervention in Eq. (5) and DPI $\rho \geq 1$ alone, and does not rely on conditions over unobserved bias like Prop. 1. The only untestable requirement of $\mathcal{T}$-invariance is later relaxed to $(\varepsilon, \mathcal{T})$-invariance to allow for sensitivity analysis over ε-misspecification of the symmetry.

C.2 BETTER BOUNDS WITH DATA AUGMENTATION

Having established validity, we now show DA can make bounds *better*, even under relaxed conditions.

Sharpness. While the centroid drift in Prop. 1 hinges on the bias direction, the identified set's *size* depends on *global* distribution shifts. This yields a milder, data-observable improvement condition:

Proposition 2 (sharper bounds with DA)**.** *Denoting* $L_2(X)$*-volume by* $|\cdot|$*, under Asms. 1b–1c and 2,*

$$\frac{|\mathcal{H}_{\widetilde{\text{pi}}}|}{|\mathcal{H}_{\text{pi}}|} \leq \Big(\rho \cdot \mathrm{tr}(\mathcal{S})/\,k\Big)^{k/2}, \qquad \frac{\mathbb{E}\big[|\mathcal{H}_{\widetilde{\text{pi}}}(X)|^2\big]}{\mathbb{E}\big[|\mathcal{H}_{\text{pi}}(X)|^2\big]} = \underbrace{\rho}_{\textit{price}} \cdot \mathrm{tr}\Big(\underbrace{\mathcal{K}}_{\textit{lever}} + \underbrace{\delta^*\delta}_{\textit{penalty}}\Big) \Big/ k.$$

Proposition 2 ties sharpening to competing factors. Noise inflation $\rho \geq 1$ and distribution shift $\mathrm{tr}(\delta^*\delta) \geq 0$ strictly expand the set. Thus, bounds tighten when variance inflation drives $\mathrm{tr}(\mathcal{S})$ small enough to satisfy $\rho\,\mathrm{tr}(\mathcal{S})/k \leq 1$. Intuitively, Lem. 2 enforces a *variance cap* on allowable bias. DA can stretch admissible candidates beyond this fixed γ-budget, pruning them from the set. Notably, the soft intervention Eq. (5) keeps the true bias $\tilde{b}$ within budget even as other candidates are pruned. Finally, as the inherited γ over-budgets the now diluted confounding by exactly ρ, one may re-calibrate to $\tilde{\gamma} := \gamma/\rho$ if the solver exposes γ, keeping validity while reducing Prop. 2's term to $\mathrm{tr}(\mathcal{S})/k$.

Informativeness. While smaller sets are desirable, we also gauge the "quality" of set candidates:

Theorem 2 (informative bounds with DA)**.** *Under Asms. 1b–1c and 2, the following holds for $d = k$,*

$$\overline{E}(\mathcal{H}_{\text{PI}}) - \overline{E}(\mathcal{H}_{\widetilde{\text{PI}}}) \quad = \quad \underbrace{E(h_{\text{erm}}) - E(h_{\widetilde{\text{erm}}})}_{\text{centroid drift (Prop. 1)}} + \frac{\sigma^2 \gamma k}{d+2} \cdot \underbrace{(1 - \rho \cdot \operatorname{tr}(\mathcal{S})/k)}_{\text{set size (Prop. 2)}} \quad =: \quad f(d),$$

And interval improvement averaged over queries: $\mathbb{E}_{X \sim p_X}\left[\overline{E}_X(\mathcal{H}_{\text{PI}}(X)) - \overline{E}_X(\mathcal{H}_{\widetilde{\text{PI}}}(X))\right] = f(1)$.

The decomposition isolates two mechanisms: the centroid drifts toward $h_\star$ and the set shrinks, as illustrated in Fig. 4b. Crucially, thresholds $\mathcal{S}[\tilde{b}] \leq 1$ and $\rho \operatorname{tr}(\mathcal{S})/k \leq 1$ are *sufficient*, and not necessary for improvement. In general, DA "wobbles" the ellipsoid of Lem. 2, inflating some directions while deflating others. The intersection $\mathcal{H}_{\text{PI}}(\boldsymbol{x}) \cap \mathcal{H}_{\widetilde{\text{PI}}}(\boldsymbol{x})$ nonetheless guarantees no regression.

Robustness. DA encoding misspecified $\mathcal{T}$-inv. induces endogenous noise $\eta \coloneqq h_\star(X) - h_\star(\widetilde{X})$,

$$Y = h_\star(X) + U + \xi, \qquad \Longleftrightarrow \qquad Y = h_\star(\widetilde{X}) + U + \eta + \xi,$$

The post-DA intervals $\mathcal{H}_{\widetilde{\text{PI}}}(\boldsymbol{x})$ may not account for this additional confounding. Therefore, to recover validity we apply ε-padding $[a, b] \pm \varepsilon \coloneqq [a - \varepsilon, b + \varepsilon]$.[1] Baseline intersection then absorbs any slack.

Theorem 3.A (robust bounds with DA)**.** *Under Asms. 1a–1b and 2, let valid baseline set $\mathcal{H}_{\text{PI}}$ cover an $(\varepsilon, \mathcal{T})$-invariant $h_\star$. Then, ε-padded post-DA intervals $\mathcal{H}_{\widetilde{\text{PI}}}(\boldsymbol{x}) \pm \varepsilon$ also remain valid on p_X-a.e. $\boldsymbol{x}$.*

Remark 1 (dominant bounds with DA)**.** Under Asms. 1a–1b and 2, let $\mathcal{H}_{\text{PI}}$ cover $(\varepsilon, \mathcal{T})$-invariant $h_\star$. Then, $\mathcal{H}_{\text{PI}}(\boldsymbol{x}) \cap (\mathcal{H}_{\widetilde{\text{PI}}}(\boldsymbol{x}) \pm \varepsilon)$ dominates $\mathcal{H}_{\text{PI}}(\boldsymbol{x})$ on sharpness and worst-case error for p_X-a.e. $\boldsymbol{x}$.

Theorem 3.A now enables DA based sensitivity analysis via the parameter ε; validity is preserved by ε-padding bounds, while baseline intersection maintains sharpness and informativeness. This is depicted in Fig. 4d where post-DA bounds may arbitrarily change, yet intersection guarantees dominance. It is worth contrasting this with §4; unlike $\mathcal{H}_{\text{PI+INV}}(\boldsymbol{x})$, $\mathcal{H}_{\widetilde{\text{PI}}}(\boldsymbol{x}) \pm \varepsilon$ solves the entire ε-grid simultaneously.

[1] For non-interval $\mathcal{H}_{\widetilde{\text{PI}}}(\boldsymbol{x})$, read $\pm\varepsilon$ as the Minkowski sum $\mathcal{H}_{\widetilde{\text{PI}}}(\boldsymbol{x}) \pm \varepsilon \coloneqq \mathcal{H}_{\widetilde{\text{PI}}}(\boldsymbol{x}) + [-\varepsilon, \varepsilon]$.

D SYMMETRY CONSTRAINT WITH T-AS-IV

A leaky IV model. The IV problem in Eq. (3) may be incomplete [Ailer et al., 2024], or even misspecified [Watson et al., 2024, Penn et al., 2025, Conley et al., 2012, Deaner, 2022, 2025, Cinelli and Hazlett, 2025]. Consequently, $h_\star$ is unidentifiable. Nevertheless, relaxed IV assumptions are still useful for sharpening PI bounds Balke and Pearl [1997], Kilbertus et al. [2020]:

Assumption 3 (a leaky IV sensitivity model Deaner [2022])**.** For exclusion/ un-confoundedness budget $\gamma_{\boldsymbol{z}}$, let

$$\mathcal{P}_{\text{PI+IV}} \coloneqq \left\{ q \in \mathcal{P}_{\text{PI}} \,:\, \left\| \mathbb{E}_p[Y - h_\star^q(X) \,|\, Z] \right\|_Z^2 \leq \sigma^2 \gamma_{\boldsymbol{z}} \right\}.$$

Here, $\gamma_{\boldsymbol{z}}$ represents a bounded violation in un-confoundedness and/or exclusion of Z, while the baseline constraints $\mathcal{P}_{\text{PI}}$ may allow $\mathcal{P}_{\text{PI+IV}}$ to be sufficiently well-posed to yield non-vacuous bounds.

Note that post-DA, the random transformation T satisfies the IV properties of unconfoundedness $T \perp\!\!\!\perp U$ and exclusion $Y \perp\!\!\!\perp T \,|\, X, U$ by construction. However, $\mathcal{T}$-inv. renders it incomplete; since $T \perp\!\!\!\perp X$, so $\mathbb{E}[h_\star(\widetilde{X}) \,|\, T] = \mathbb{E}[h_\star(X) \,|\, T] = \mathbb{E}[h_\star(X)] = 0$. Nevertheless, following how incomplete instruments are still useful to constrain PI, we exploit this connection to enforce symmetry within off-the-shelf, or black-box IV-based PI solvers by plugging in T as an instrument $\widetilde{Z} \coloneqq (Z, T)$,

$$\mathcal{Q}_{\widetilde{\text{PI}}+\widetilde{\text{IV}}}(p_{XYZ}) \coloneqq \mathcal{Q}_{\text{PI+IV}}(p_{\widetilde{X}Y\widetilde{Z}}).$$

We defer readers to §E for a comparison of Asm. 3 with §4.

Misspecified DA under IV constraints. For completeness, we explore the limits of the ε-padding approach from Thm. 3.A under the IV constraint in Asm. 3. Notice that $\mathcal{T}$-invariance violation manifests as violated IV exclusion / un-confoundedness. This violation may be accommodated by either DA reducing confounding via the DPI $\sigma^2 \leq \tilde{\sigma}^2$, or a sufficient slack in the baseline IV budget $\gamma_{\boldsymbol{z}}$.

Theorem 3.B (robust bounds with DA)**.** *Let a valid model following Asm. 3 yield intervals $\mathcal{H}_{\text{PI+IV}}(\boldsymbol{x})$ under the setting of Thm. 3.A. Then, $\mathcal{H}_{\widetilde{\text{PI}}+\widetilde{\text{IV}}}(\boldsymbol{x}) \pm \varepsilon$ is also valid for $\widetilde{Z} \in \{Z, (Z, T)\}$ on p_X-a.e. $\boldsymbol{x}$ if*

$$\varepsilon \leq \tilde{\sigma}\sqrt{\gamma_{\boldsymbol{z}}} - \sigma\sqrt{\gamma_{\boldsymbol{z}}^\star}, \qquad \text{where } \gamma_{\boldsymbol{z}}^\star \leq \gamma_{\boldsymbol{z}} \text{ is the true IV budget.}$$

The above inequality allows for some tolerance to ε-misspecified symmetry under DA so long as $\gamma_{\boldsymbol{z}}$ is positive. In practice this is often true even for perfect instruments to make the optimization problem more tractable. Importantly, practitioners can use the following observable diagnostic for Thm. 3.B:

$$\varepsilon \leq (\tilde{\sigma} - \sigma)\sqrt{\gamma_{\boldsymbol{z}}}.$$

Alternatively, if the sensitivity parameter $\gamma_{\boldsymbol{z}}$ is exposed, one can simply increase it to $\tilde{\gamma}_{\boldsymbol{z}}$ satisfying

$$\tilde{\sigma}\sqrt{\tilde{\gamma}_{\boldsymbol{z}}} \geq \varepsilon + \sigma\sqrt{\gamma_{\boldsymbol{z}}}.$$

Meaning the new budget $\tilde{\gamma}_{\boldsymbol{z}}$ should accommodate the IV as well as the symmetry misspecification.

E SHAPE CONSTRAINT VS. CHANGE OF MEASURE

In this section we aim to compare the two operationalizations of symmetry informed PI; via an invariance shape constraint in §4, and via change-of-measure through DA pre-processing in §5. Throughout, we assume no observable co-variates $Z = \varnothing$ for simplicity, and a mean-centered Y w.l.o.g. with the mean absorbed into the intercept of $h_\star$. Define the ambient $\mathcal{T}$-inv. subspace $\mathcal{I}$ as

$$\mathcal{I} \coloneqq \left\{ h \in L_2^0(X) \cap L_2^0(\widetilde{X}) \;:\; h(X) = h(\widetilde{X}) \text{ a.s.} \right\}, \qquad \mathcal{H}_{\text{inv}} \coloneqq \mathcal{H} \cap \mathcal{I}.$$

Notice that, by construction, all forms $\langle\cdot,\cdot\rangle_X$, $\langle\cdot,\cdot\rangle_{\widetilde{X}}$ $\langle\cdot,\cdot\rangle_{L_2(X)}$ and $\langle\cdot,\cdot\rangle_{L_2(\widetilde{X})}$ coincide on $\mathcal{I}, \mathcal{H}_{\text{inv}}$. Next, define the operator $\mathcal{E}_{A|B} : L_2^0(A) \to L_2^0(B)$ as conditional expectation on centered subspaces

$$(\mathcal{E}_{A|B}\, h)(\cdot) = \mathbb{E}[h(A) \,|\, B = \cdot].$$

Well defined because $\mathbb{E}[\mathbb{E}[h(A) \,|\, B]] = \mathbb{E}[h(A)] = 0$ by the tower property, and $\mathcal{E}_{A|B} h = 0$ iff $\mathbb{E}[h(A) \,|\, B] = 0$ a.s.. Of particular interest are $\mathcal{E}_{X|\widetilde{X}}$ and $\mathcal{E}_{\widetilde{X}|T}$, both of which we discuss below.

Now, under a $\mathcal{T}$-inv. $h_\star$, $\varepsilon = 0$ and the invariance shape constraint in §4 yields $\mathcal{H}_{\text{pi+inv}} = \mathcal{H}_{\text{pi}} \cap \mathcal{H}_{\text{inv}}$ (see Cor. 1). By contrast, DA pre-processing yields $\mathcal{H}_{\widetilde{\text{pi}}}$ and/or $\mathcal{H}_{\widetilde{\text{pi}}+\widetilde{\text{iv}}}$. To make their comparison tractable in observables, we make two standard assumptions. First, we take $\mathcal{T}$ to be a compact group:

Assumption 4 (compact group action DA)**.** Let $\mathcal{T}$ be a compact group with action $\alpha : \mathcal{X} \times \mathcal{T} \to \mathcal{X}$. Furthermore, $\mathcal{T}$ is endowed with a (unique) normalized Haar measure p_T, i.e. p_T is uniform over $\mathcal{T}$.

Assumption 4 encompasses several standard DA transformations, including some used in our experiments such as rotations, translations, etc. Under Asm. 4, $p_{\widetilde{X}}$ itself becomes $\mathcal{T}$-invariant. Importantly, $\mathcal{E}_{X|\widetilde{X}}$ is now the Reynolds operator; an $L_2(X)$-orthogonal projector onto $\mathcal{I}$ Rota [1964]. Under Asms. 1a–1b, the post-DA bias $\tilde{b}$ is then a projection of b onto $\mathcal{I}$ (Lem. 4). This now allows us to compare identification metrics entirely in observables, e.g. $\tilde{b} \in \mathcal{H}_{\text{inv}}$ reduces $\mathcal{S}[\tilde{b}] = \mathcal{K}[\tilde{b}] + (\delta^*\delta)[\tilde{b}] = 1$ in Prop. 1.

Second, since T-as-IV imposes Eq. (3) as a constraint, it filters functions with $\mathbb{E}[h(\widetilde{X})|T]$ a.s. constant. This holds for all $h \in \mathcal{H}_{\text{inv}}$, since $\mathbb{E}[h(\widetilde{X})|T] = \mathbb{E}[h(X)|T] = \mathbb{E}[h(X)] = 0$ a.s. following $T \perp\!\!\!\perp X$. Hence $\mathcal{H}_{\text{inv}} \subseteq \text{null}\big(\mathcal{E}_{\widetilde{X}|T}\big)$. We say that DA is *relevant*, or *complete*, when this holds with equality:

Assumption 5 (relevant/complete DA)**.** Let T be *relevant / complete*: $\qquad \text{null}\big(\mathcal{E}_{\widetilde{X}|T}\,\big|_{\mathcal{H}_{\widetilde{X}}}\big) = \mathcal{H}_{\text{inv}}.$

Assumption 5 is akin to the standard injective $\mathcal{E}_{X|Z}$ assumption for relevant/complete IVs Z Ailer et al. [2024], Sun et al. [2025]. Similarly, Asm. 5 assumes $\mathcal{E}_{\widetilde{X}|T}$ injective on $\widetilde{X}$-orthogonal complement of $\mathcal{H}_{\text{inv}}$ within $\mathcal{H}_{\widetilde{X}}$. Meaning, the only directions whose data-averaged response is zero across augmentations are the invariant ones.

The inv. shape constraint and change-of-measure routes are now comparable under Asms. 4 and 5:

Proposition 3 (shape constraint vs. change-of-measure)**.** *Under Asms. 1a–1b, and Asms. 2 to 5 with $\mathcal{T}$-invariant $h_\star$ and $Z = \varnothing$. Set $\varepsilon = \gamma_{\boldsymbol{z}} = 0$, and suppose the baseline PI set is valid $h_\star \in \mathcal{H}_{\text{pi}}$. Then, $h_{\widetilde{\text{erm}}} = \mathcal{T} h_{\text{erm}} \in \mathcal{H}_{\text{inv}}$, where $\mathcal{T}$ is the $L_2(X)$-orthogonal projection onto $\mathcal{I}$ (cf. §H). Further,*

$$\mathcal{H}_{\widetilde{\text{pi}}} = \left\{ h \in \mathcal{H}_{\widetilde{X}} \;:\; \left\| h - h_{\widetilde{\text{erm}}} \right\|_{\widetilde{X}}^2 \le \tilde{\sigma}^2 \gamma \right\},$$
$$\mathcal{H}_{\widetilde{\text{pi}}+\widetilde{\text{iv}}} = \left\{ h \in \mathcal{H}_{\text{inv}} : \left\| h - h_{\widetilde{\text{erm}}} \right\|_{\widetilde{X}}^2 \le \tilde{\sigma}^2 \gamma \right\},$$
$$\mathcal{H}_{\text{pi+inv}} = \left\{ h \in \mathcal{H}_{\text{inv}} : \left\| h - h_{\widetilde{\text{erm}}} \right\|_{X}^2 \le \sigma^2 (1 + \gamma - \rho) \right\}.$$

All identified sets are balls centered about $h_{\widetilde{\text{erm}}}$. Both $\mathcal{H}_{\widetilde{\text{pi}}+\widetilde{\text{iv}}}$ and $\mathcal{H}_{\text{pi+inv}}$ live on the slice $\mathcal{H}_{\text{inv}}$, the latter with a smaller radius $\sigma\sqrt{1+\gamma-\rho} \le \tilde{\sigma}\sqrt{\gamma}$ with equality iff $\rho = 1$. It trivially follows that,

$$h_\star \in \mathcal{H}_{\text{pi+inv}} \subseteq \mathcal{H}_{\widetilde{\text{pi}}+\widetilde{\text{iv}}} \subseteq \mathcal{H}_{\widetilde{\text{pi}}},$$

and the same dominance hierarchy holds for sharpness and worst-case errors over $\mathcal{H}_\bullet$ and $\mathcal{H}_\bullet(\boldsymbol{x})$.

Under Asm. 1c, closed form volume, width and avg. errors for the balls in Prop. 3 can now be derived in observables similar to Prop. 2 and Thm. 2. We leave it as an exercise for interested readers.

Algorithm 1 Invariance shape constraint (§4)

Input: Dataset $\mathcal{D}$; transformation set $\mathcal{T}$ with measure p_T; augmentation folds m;
structural constraints $\mathcal{P}_{\text{PI}}, \mathcal{H}$; test functions $\mathcal{F}$; tolerances ε, γ_0; query point $\boldsymbol{x}$.
$\mathcal{H}_{\text{PI+INV}}(\boldsymbol{x}) \leftarrow$ solve $\min$ / $\max$ $h^q_\star(\boldsymbol{x})$ in (P1) using $\mathcal{D}$ and $c = 1$.
Output: $\mathcal{H}_{\text{PI+INV}}(\boldsymbol{x})$.

Algorithm 2 Change of measure via DA pre-processing (§5)

Input: Dataset $\mathcal{D}$; transformation set $\mathcal{T}$ with measure p_T; augmentation folds m;
structural constraints $\mathcal{P}_{\text{PI}}, \mathcal{H}$; test functions $\mathcal{F}$; tolerances ε, γ_0; query point $\boldsymbol{x}$.
Take m samples $\boldsymbol{\tau}_{i,j} \sim p_T$ for each $i \in [n]$ with $n = |\mathcal{D}|$.
Construct dataset $\widetilde{\mathcal{D}} := \{(\tilde{\boldsymbol{x}}_{i,j},\, y_i,\, \tilde{\boldsymbol{z}}_{i,j})\}_{i\in[n],j\in[m]}$ with $\tilde{\boldsymbol{x}}_{i,j} := \boldsymbol{\tau}_{i,j}\boldsymbol{x}_i$ and $\tilde{\boldsymbol{z}}_{i,j} := (\boldsymbol{z}_i, \boldsymbol{\tau}_{i,j})$.
$\mathcal{H}_{\text{PI}}(\boldsymbol{x}) \leftarrow$ solve $\min$ / $\max$ $h^q_\star(\boldsymbol{x})$ in (P1) using $\mathcal{D}$ and $c = 0$.
$\mathcal{H}_{\widetilde{\text{PI}}}(\boldsymbol{x}) \leftarrow$ solve $\min$ / $\max$ $h^q_\star(\boldsymbol{x})$ in (P1) using $\widetilde{\mathcal{D}}$ and $c = 0$.
Output: $\mathcal{H}_{\text{PI}}(\boldsymbol{x}) \cap \big(\mathcal{H}_{\widetilde{\text{PI}}}(\boldsymbol{x}) \pm \varepsilon\big)$.

F IMPLEMENTATION DETAILS

We first write the finite-n sample version of the abstract partial identification program from §1 as:

$$\begin{aligned} &\min_q \,/\, \max_q\ h^q_\star(\boldsymbol{x}) && \text{// interval lower / upper bounds} \\ &\quad \text{s.t.}\quad D_{\mathcal{F}}(q_{XYZ},\, \widehat{p}_{XYZ}) \le \gamma_n, && \text{// observational consistency} \\ &\qquad q \in \mathcal{P}_{\text{PI}}, \quad h^q_\star \in \mathcal{H}, \quad c \cdot \widehat{E}_{\text{INV}}(h^q_\star) \le \varepsilon^2. && \text{// structural constraints} \end{aligned} \tag{P1}$$

Here, $\widehat{p}_{XYZ}$ is the empirical distribution of n observations. The IPM $D_{\mathcal{F}}$ matches q_{XYZ} to $\widehat{p}_{XYZ}$ within tolerance $\gamma_n \ge 0$ to account for sampling error. Further, $\widehat{E}_{\text{INV}}$ is a Monte Carlo approx. of E_{INV}:

$$\widehat{E}_{\text{INV}}(h) := \frac{1}{mn}\sum_{i=1}^{n}\sum_{j=1}^{m} \big|h(\boldsymbol{x}_i) - h(\boldsymbol{\tau}_{i,j}\,\boldsymbol{x}_i)\big|^2, \qquad (\boldsymbol{x}_i, \cdot) \in \mathcal{D}, \quad \boldsymbol{\tau}_{i,j} \sim p_T.$$

In practice, the partial-r^2 constraints (Asm. 2) directly impose the ellipsoid of Lem. 2: $\widehat{E}_{\text{R2}}(h) \le \widehat{\sigma}^2\gamma$,

$$\widehat{E}_{\text{R2}}(h) := \frac{1}{n}\sum_{i=1}^{n}\big(h(\boldsymbol{x}_i) - \widehat{h}_{\text{ERM}}(\boldsymbol{x}_i)\big)^2, \qquad \widehat{\sigma}^2 := \frac{1}{n}\sum_{i=1}^{n}\big(y_i - \widehat{h}_{\text{ERM}}(\boldsymbol{x}_i)\big)^2.$$

For the leaky-IV constraint of Asm. 3, we implement a generalized method of moments (GMM) relaxation of the conditional moment restriction Akbar et al. [2025], Ailer et al. [2024]. For IV test functions $\phi_{\boldsymbol{z}} : \mathcal{Z} \to \mathbb{R}^{d_{\phi_{\boldsymbol{z}}}}$, define

$$\widehat{R}_{\text{IV}}(h) := \left\| \frac{1}{n}\sum_{i=1}^{n}\big(y_i - h(\boldsymbol{x}_i)\big)\cdot\phi_{\boldsymbol{z}}(\boldsymbol{z}_i) \right\|^2_{\boldsymbol{W}}, \qquad \|\boldsymbol{v}\|^2_{\boldsymbol{W}} := \boldsymbol{v}^\top \boldsymbol{W}\boldsymbol{v}.$$

We use $\phi_{\boldsymbol{z}}(\boldsymbol{z}) = \boldsymbol{z}$ and $\boldsymbol{W} = \widehat{\boldsymbol{\Sigma}}^{\dagger}_{\boldsymbol{z}}$ Bennett et al. [2019], where $\dagger$ denotes the Moore–Penrose pseudoinverse. For point-estimation $\widehat{h}_{\text{IV}} := \operatorname{argmin}_{h\in\mathcal{H}} \widehat{R}_{\text{IV}}(h)$, whereas for PI (relaxed Asm. 3) we constrain $\widehat{R}_{\text{IV}}(h) \le \widehat{\sigma}^2\gamma_{\boldsymbol{z}}$.

We instantiate a parameterized hypothesis class $\mathcal{H}_{\boldsymbol{\theta}} := \{h_{\boldsymbol{\theta}} : \boldsymbol{\theta} \in \Theta\}$. With $\gamma_n = 0$, (P1) under Asms. 2 and 3 reduces to:

$$\begin{aligned} &\min_{\boldsymbol{\theta}\in\Theta} \,/\, \max_{\boldsymbol{\theta}\in\Theta}\ h_{\boldsymbol{\theta}}(\boldsymbol{x}) \\ &\quad \text{s.t.}\quad \widehat{E}_{\text{R2}}(h_{\boldsymbol{\theta}}) \le \widehat{\sigma}^2\gamma, && \text{// bounded confounding (Lem. 2)} \\ &\qquad \widehat{R}_{\text{IV}}(h_{\boldsymbol{\theta}}) \le \widehat{\sigma}^2\gamma_{\boldsymbol{z}}, && \text{// leaky IV constraints (Asm. 3)} \\ &\qquad c\cdot\widehat{E}_{\text{INV}}(h_{\boldsymbol{\theta}}) \le \varepsilon^2, && \text{// invariance constraint (§4)} \\ &\qquad \textstyle\sum_i h_{\boldsymbol{\theta}}(\boldsymbol{x}_i) = \sum_i y_i. && \text{// centering/mean-matching} \end{aligned} \tag{P2}$$

Algos. 1 and 2 list how programs (P1) and (P2) are invoked by different approaches to incorporate known invariance in PI.

G EXPERIMENT SUPPLEMENT

G.1 SIMULATION EXPERIMENT

Fix $\boldsymbol{\theta}_\star \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$ and sample a $\boldsymbol{\Sigma}_{\boldsymbol{x}u}$ uniformly on a unit sphere. Generate

$$h_\star(\boldsymbol{x}) \coloneqq \boldsymbol{\theta}_\star^\top \boldsymbol{x}, \qquad (X, U) \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} \boldsymbol{I} & \boldsymbol{\Sigma}_{\boldsymbol{x}u} \\ \boldsymbol{\Sigma}_{\boldsymbol{x}u}^\top & 1 \end{bmatrix}\right), \qquad \xi \sim \mathcal{N}(0, 0.1^2),$$

and Y from Eq. (1). Next, we follow Akbar et al. [2025] to construct T such that $h_\star$ respects $\mathcal{T}$-inv. By selecting $d_{\boldsymbol{\tau}}$ basis of $\mathrm{null}(\boldsymbol{\theta}_\star)$, each with probability 0.8, as rows of $\boldsymbol{A} \in \mathbb{R}^{d_{\boldsymbol{\tau}} \times d_{\boldsymbol{x}}}$, we define:

$$\widetilde{X} = \alpha(X, T) = X + \boldsymbol{A}^\top T, \qquad T \sim \mathcal{N}(\mathbf{0}_{d_{\boldsymbol{\tau}}}, \boldsymbol{I}_{d_{\boldsymbol{\tau}}}).$$

By construction, $\widetilde{X}$ varies only in the null-space of $\boldsymbol{\theta}_\star$, ensuring $h_\star$ is $\mathcal{T}$-invariant. While this construction utilizes the true $\boldsymbol{\theta}_\star$, we treat $\boldsymbol{A}$ as illustrative of domain knowledge defining plausible data symmetries; crucially, this knowledge alone is insufficient to identify $\boldsymbol{\theta}_\star$ without observing U.

Taking dimensions $d_{\boldsymbol{x}} = 32$, $d_{\boldsymbol{\tau}} = 31$, we generate $n = 2048$ samples for (X, Y, T). The ERM baseline is the closed-form OLS solution. And for PI we use the partial-r^2 model (Asm. 2) with the effective confounding budget $\gamma = 1$, and invariance-error and IV validity budgets of $\varepsilon = \gamma_{\boldsymbol{z}} = 2^{-8}$.

Figure 5 *(Left, Right)* displays the identified intervals $\mathcal{H}_{\mathrm{pi}}(\boldsymbol{x})$ and $\mathcal{H}_{\widetilde{\mathrm{pi}}}(\boldsymbol{x})$ over query points $\boldsymbol{x} \coloneqq t \cdot \boldsymbol{\nu}_i$ along the first two principal components $\boldsymbol{\nu}_1, \boldsymbol{\nu}_2$, and sweeping t across ± 3 standard deviations s_i. Figure 5 *(Center)* shows a radial sweep $\boldsymbol{x} \coloneqq s_1 \cdot \sin(\vartheta) \cdot \boldsymbol{\nu}_1 + s_2 \cdot \cos(\vartheta) \cdot \boldsymbol{\nu}_2$ for $\vartheta \in [0, 2\pi]$, demonstrating consistent interval contraction across directions.

G.2 OPTICAL DEVICE EXPERIMENT

We utilize the benchmark dataset by Janzing and Schölkopf [2018], consisting of 3×3 pixel images X displayed on a laptop screen which generate voltage readings Y across a photo-diode. The system involves a physically instantiated latent confounder U controlling the intensity of two LEDs; the first affects the webcam capturing X, while the second influences the photo-diode measuring Y. We estimate the true target $h_\star(\boldsymbol{x}) = \boldsymbol{\theta}_\star^\top \phi(\boldsymbol{x})$ via back-door adjustment using joint features $(\phi(X), U)$, where $\phi(X)$ denotes polynomial features of X. We select the polynomial degree $\deg(h_\star) \in \{1, \cdots, 5\}$ that best explains the data ($\deg(h_\star) = 2$ for Fig. 6). Our choice of DA on X composes random vertical/horizontal flips and $90°$ rotations. For a dataset of $n = 1000$ samples, we compute features $\phi(\widetilde{X})$ to be used with PI, setting effective budget $\gamma = 2^{-2}$ for the partial-r^2 model, and $\varepsilon = 2^{-1.5}, \gamma_{\boldsymbol{z}} = 2^{-6}$ for the inv. error and IV budgets respectively. Using the visualization approach of Fig. 5, Fig. 6 shows that symmetry information does improve PI.

H PROOFS

COMMON SETUP

Finite dimension. For $V \in \{X, \widetilde{X}\}$, define the centering operator $\mathcal{C}_V \coloneqq h - \mathbb{E}[h(V)]$. Now, since $\mathcal{H} + \mathbb{R}1 = \mathcal{H}$ (Asm. 1), therefore $\mathcal{C}_V h \in \mathcal{H}_V$. These maps define a bijection between $\mathcal{H}_X$ and $\mathcal{H}_{\widetilde{X}}$,

$$\mathcal{H}_X \underset{\mathcal{C}_X}{\overset{\mathcal{C}_{\tilde{X}}}{\rightleftarrows}} \mathcal{H}_{\widetilde{X}}$$

therefore, $\dim(\mathcal{H}_{\widetilde{X}}) = \dim(\mathcal{H}_X) = k$ (Asm. 1c).

Non-degeneracy. $p_X \ll p_{\widetilde{X}}$ (Asm. 1b) transfers Hilbert structure (Asm. 1c) to $\mathcal{H}_{\widetilde{X}}$. For $h \in \mathcal{H}_{\widetilde{X}}$,

$$\begin{aligned} \|h\|_{\widetilde{X}} = 0 \quad &\implies \quad h(\widetilde{X}) = 0 \text{ a.s.} && \text{(0 is the only a.s. constant } h \in \mathcal{H}_{\widetilde{X}}.\text{)} \\ &\overset{1b}{\implies} \quad h(X) = 0 \text{ a.s.} \quad \implies \quad \|h\|_X = 0 \quad \overset{1c}{\implies} \quad h \in \mathbb{R}1 \quad \overset{h \in \mathcal{H}_{\widetilde{X}}}{\implies} \quad h = 0. \end{aligned}$$

Hence both $\langle \cdot, \cdot \rangle_X$ and $\langle \cdot, \cdot \rangle_{\widetilde{X}}$ are positive-definite Hilbert norms on $\mathcal{H}_{\widetilde{X}}$, but so is $\langle \cdot, \cdot \rangle_{L_2(X)}$ because

$$\|h\|^2_{L_2(X)} = \|h\|^2_X + (\delta h)^2 \geq \|h\|^2_X.$$

Invariant slice. Recall from §E,

$$\mathcal{I} \coloneqq \left\{ f \in L_2^0(X) \cap L_2^0(\widetilde{X}) \;\; : \;\; f(X) = f(\widetilde{X}) \text{ a.s.} \right\}.$$

Here $\mathcal{I}$ is a closed linear subspace, irrespective of any linearity assumed of $\mathcal{H}$. Now define

$$\mathcal{H}_{\text{inv}} \coloneqq \mathcal{H} \cap \mathcal{I}.$$

By construction, all four forms coincide on $\mathcal{I}$ and $\mathcal{H}_{\text{inv}}$,

$$\langle f, g \rangle_{L_2(X)} \;=\; \langle f, g \rangle_{L_2(\widetilde{X})} \;=\; \langle f, g \rangle_X \;=\; \langle f, g \rangle_{\widetilde{X}}, \qquad \forall\, f, g \in \mathcal{I}. \tag{6}$$

Covariance forms agree with L_2 as $\mathcal{I}$ is both X- and $\widetilde{X}$-centered. And X-, $\widetilde{X}$-forms agree by $\mathcal{T}$-inv.

Throughout, we denote $\mathcal{\pi}$ as the $L_2(X)$-orthogonal (equivalently, X-orthogonal) projection onto $\mathcal{I}$.

H.1 PROOF OF THM. 1—VALID BOUNDS WITH DA

Theorem 1 (valid bounds with DA)**.** *Let $h_\star$ be $\mathcal{T}$-inv. under Asm. 1a and 2. Then, the post-DA set $\mathcal{H}_{\widetilde{\text{PI}}}$ is exactly valid whenever the baseline set $\mathcal{H}_{\text{PI}}$ is ϵ-valid with $\epsilon \le \sigma^2\big(\sqrt{\rho - 1 + \gamma\rho} - \sqrt{\gamma}\big)^2$.*

Proof. We proceed as: (i) express baseline approx. error as point-to-ball distance ($*$), (ii) certify post-DA membership ($\dagger$) by comparing bias magnitudes ($\ddagger$) using Prop. 4.

First, write the Lem. 2 pre- and post-DA partially identified sets in their respective L_2 geometries as:

$$\mathcal{H}_{\text{PI}} = \Big\{h \in \mathcal{H}_X : \|h - h_{\text{erm}}\|^2_{L_2(X)} \le \gamma\sigma^2\Big\},$$
$$\mathcal{H}_{\widetilde{\text{PI}}} = \Big\{h \in \mathcal{H}_{\widetilde{X}} : \|h - h_{\widetilde{\text{erm}}}\|^2_{L_2(\widetilde{X})} \le \gamma\tilde{\sigma}^2\Big\}.$$

Note that target $h_\star$ lies in both centered classes: $\mathbb{E}[h_\star(X)] = \mathbb{E}[Y] = 0 \implies h_\star \in \mathcal{H}_X$, and by $\mathcal{T}$-inv., $\mathbb{E}[h_\star(\widetilde{X})] = \mathbb{E}[h_\star(X)] = 0 \implies h_\star \in \mathcal{H}_{\widetilde{X}}$. Define the pre- and post-DA bias magnitudes:

$$A \coloneqq \|h_{\text{erm}} - h_\star\|_{L_2(X)}, \qquad B \coloneqq \|h_{\widetilde{\text{erm}}} - h_\star\|_{L_2(\widetilde{X})}.$$

For an $L_2(X)$-round ball, approx. error $E^-(\bullet) = \min_{h\in\bullet} \Big\|h - h_\star\Big\|^2_{L_2(X)}$ reduces to point-to-ball distance. Since $\mathcal{H}_{\text{PI}}$ is essentially $\mathcal{H}_X \cap L_2(X)$-ball, approx. error is lower-bounded by point-to-ball distance,

$$E^-(\mathcal{H}_{\text{PI}}) \ge \big(A - \sigma\sqrt{\gamma}\big)^2_+, \qquad h_\star \in \mathcal{H}_{\text{PI}} \iff A \le \sigma\sqrt{\gamma}. \tag{$*$}$$

As for $\mathcal{H}_{\widetilde{\text{PI}}}$, the following validity criterion gives $E^-(\mathcal{H}_{\widetilde{\text{PI}}}) = 0$ exactly,

$$h_\star \in \mathcal{H}_{\widetilde{\text{PI}}} \iff B \le \tilde{\sigma}\sqrt{\gamma}. \tag{$\dagger$}$$

For $C^2 \coloneqq \tilde{\sigma}^2 - \sigma^2$, Prop. 4 (Asm. 1a, $\mathcal{T}$-inv. $h_\star$) supplies the following two identities with $C^2 \ge 0$,

$$A^2 = B^2 + C^2, \qquad \tilde{\sigma}^2 = \sigma^2 + C^2. \tag{$\ddagger$}$$

Now suppose $\mathcal{H}_{\text{PI}}$ is ϵ-valid with

$$\epsilon \le \epsilon^+ \coloneqq \Big(\sqrt{C^2 + \gamma\tilde{\sigma}^2} - \sigma\sqrt{\gamma}\Big)^2 = \sigma^2\Big(\sqrt{\rho - 1 + \gamma\rho} - \sqrt{\gamma}\Big)^2,$$

the second equality substituting $C^2 + \gamma\tilde{\sigma}^2 = \sigma^2(\rho - 1 + \gamma\rho)$ from Eq. ($\ddagger$). Note $\epsilon^+ > 0$ iff $\rho > 1$, and

$$\sqrt{\epsilon^+} = \sqrt{C^2 + \gamma\tilde{\sigma}^2} - \sigma\sqrt{\gamma} \ge 0,$$

since the difference defining ϵ^+ is non-negative $C^2 + \gamma\tilde{\sigma}^2 - \gamma\sigma^2 = (1+\gamma)C^2 \ge 0$. From Eq. ($*$),

$$\begin{aligned}
&(A - \sigma\sqrt{\gamma})_+ \le \sqrt{\epsilon}, && (\epsilon\text{-validity.})\\
\implies\quad &A - \sigma\sqrt{\gamma} \le \sqrt{\epsilon^+}, && (x \le (x)_+ \text{ and } \sqrt{\epsilon} \le \sqrt{\epsilon^+}.)\\
\implies\quad &A - \sigma\sqrt{\gamma} \le \sqrt{C^2 + \gamma\tilde{\sigma}^2} - \sigma\sqrt{\gamma}, && (\text{Def. of } \epsilon^+, \text{ and } \sqrt{\epsilon^+} \ge 0.)\\
\implies\quad &A^2 \le C^2 + \gamma\tilde{\sigma}^2. && (\text{Add } \sigma\sqrt{\gamma}, \text{ then square.})\\
\implies\quad &B^2 = A^2 - C^2 \le \gamma\tilde{\sigma}^2. && (\text{From Eq. } (\ddagger).)
\end{aligned}$$

Because $B \le \tilde{\sigma}\sqrt{\gamma}$, thus by Eq. ($\dagger$) $h_\star \in \mathcal{H}_{\widetilde{\text{PI}}}$, and therefore

$$E^-(\mathcal{H}_{\widetilde{\text{PI}}}) = 0 \le E^-(\mathcal{H}_{\text{PI}}).$$

Finally, equality holds precisely when $E^-(\mathcal{H}_{\text{PI}}) = 0$, which by Eq. ($*$) is equivalent to $h_\star \in \mathcal{H}_{\text{PI}}$. □

H.2 PROOF OF THM. 2—INFORMATIVE BOUNDS WITH DA

Theorem 2 (informative bounds with DA)**.** *Under Asms. 1b–1c and 2, the following holds for $d = k$,*

$$\overline{E}(\mathcal{H}_{\text{PI}}) - \overline{E}(\mathcal{H}_{\widetilde{\text{PI}}}) \quad = \quad \underbrace{E(h_{\text{erm}}) - E(h_{\widetilde{\text{erm}}})}_{\text{centroid drift (Prop. 1)}} + \frac{\sigma^2\gamma k}{d+2} \cdot \underbrace{(1 - \rho \cdot \operatorname{tr}(\mathcal{S})/k)}_{\text{set size (Prop. 2)}} \quad =: \quad f(d),$$

And interval improvement averaged over queries: $\mathbb{E}_{X\sim p_X}\left[\overline{E}_X(\mathcal{H}_{\text{PI}}(X)) - \overline{E}_X(\mathcal{H}_{\widetilde{\text{PI}}}(X))\right] = f(1)$.

Proof. Gist of the proof: (i) reduce avg. error to second moment of a uniform law over a centered-ball, (ii) connect it to post-DA ball via operator $\mathcal{S}$, and (iii) repeat for intervals via representers and Lem. 5.

Setup. Write the Lem. 2 and Cor. 3 pre- and post-DA PI sets in their respective L_2 geometries:

$$\mathcal{H}_{\text{PI}} = \left\{h \in \mathcal{H}_X : \|h - h_{\text{erm}}\|^2_{L_2(X)} \le \sigma^2\gamma\right\}, \qquad \mathcal{H}_{\text{PI}}(\boldsymbol{x}) = \{h_{\text{erm}}(\boldsymbol{x})\} \pm \sigma\sqrt{\gamma}\|g_{\boldsymbol{x}}\|_{L_2(X)},$$
$$\mathcal{H}_{\widetilde{\text{PI}}} = \left\{h \in \mathcal{H}_{\widetilde{X}} : \|h - h_{\widetilde{\text{erm}}}\|^2_{L_2(\widetilde{X})} \le \tilde{\sigma}^2\gamma\right\}, \qquad \mathcal{H}_{\widetilde{\text{PI}}}(\boldsymbol{x}) = \{h_{\widetilde{\text{erm}}}(\boldsymbol{x})\} \pm \tilde{\sigma}\sqrt{\gamma}\|\tilde{g}_{\boldsymbol{x}}\|_{L_2(\widetilde{X})},$$

with $L_2(X)$-representers $g_{\boldsymbol{x}}, \tilde{g}_{\boldsymbol{x}}$ well defined on Hilbert $\mathcal{H}_X$ (Asm. 1c) and $\mathcal{H}_{\widetilde{X}}$ (Asm. 1b, cf. §H). Now, suppose u is uniform on a centered d-dim. ball of radius r w.r.t. *any* inner product $\langle\cdot,\cdot\rangle$, then

$$\mathbb{E}[u] = 0, \qquad\qquad \mathbb{E}[\langle u, \mathcal{A}u\rangle] = \frac{r^2}{d+2}\operatorname{tr}(\mathcal{A}),$$

for any self-adjoint operator $\mathcal{A}$ in $\langle\cdot,\cdot\rangle$. We show that avg. errors reduce to these identities up to offset.

Set version ($d = k$)**.** Recall $\overline{E}(\cdot) = \mathbb{E}_{h\sim\text{Unif}(\cdot)}\left[\|h - h_\star\|^2_{L_2(X)}\right]$. Take the re-parameterizations

$$h = h_{\text{erm}} + u, \qquad u \sim \text{Unif}\left\{u \in \mathcal{H}_X : \|u\|_{L_2(X)} \le \sigma\sqrt{\gamma}\right\},$$
$$h = h_{\widetilde{\text{erm}}} + \tilde{u}, \qquad \tilde{u} \sim \text{Unif}\left\{u \in \mathcal{H}_{\widetilde{X}} : \|u\|_{L_2(\widetilde{X})} \le \tilde{\sigma}\sqrt{\gamma}\right\},$$

for $h \sim \text{Unif}(\mathcal{H}_{\text{PI}})$ and $h \sim \text{Unif}(\mathcal{H}_{\widetilde{\text{PI}}})$ respectively. Substituting back into avg. error $\overline{E}(\mathcal{H}_{\text{PI}})$ def.:

$$\begin{aligned}\overline{E}(\mathcal{H}_{\text{PI}}) &= \mathbb{E}_u\left[\|h_{\text{erm}} - h_\star + u\|^2_{L_2(X)}\right], \\ &= \|h_{\text{erm}} - h_\star\|^2_{L_2(X)} + \mathbb{E}_u\left[\|u\|^2_{L_2(X)}\right] = E(h_{\text{erm}}) + \frac{\sigma^2\gamma k}{k+2}, \qquad (\operatorname{tr}(\text{Id}) = k.)\end{aligned}$$

where $\mathbb{E}[u] = 0$ kills the cross-term. Likewise, for post-DA avg. error it follows from the def. of $\mathcal{S}$:

$$\begin{aligned}\overline{E}(\mathcal{H}_{\widetilde{\text{PI}}}) &= \mathbb{E}_{\tilde{u}}\left[\|h_{\widetilde{\text{erm}}} - h_\star + \tilde{u}\|^2_{L_2(X)}\right], \\ &= \|h_{\widetilde{\text{erm}}} - h_\star\|^2_{L_2(X)} + \mathbb{E}_{\tilde{u}}\left[\langle\tilde{u}, \mathcal{S}\tilde{u}\rangle_{L_2(\widetilde{X})}\right], \qquad (\|h\|^2_{L_2(X)} = \langle h, \mathcal{S}h\rangle_{L_2(\widetilde{X})} \text{ for } h \in \mathcal{H}_{\widetilde{X}}.) \\ &= E(h_{\widetilde{\text{erm}}}) + \frac{\tilde{\sigma}^2\gamma}{k+2}\operatorname{tr}(\mathcal{S}).\end{aligned}$$

Again, cross-terms vanish because $\mathbb{E}[\tilde{u}] = 0$. Subtracting, and with $\rho = \tilde{\sigma}^2/\sigma^2$ gives the result $f(k)$.

Interval version ($d = 1$)**.** A uniform draw from $[-w, w]$ has mean zero and variance $w^2/3$. Thus,

$$\overline{E}_{\boldsymbol{x}}(\mathcal{H}_{\text{PI}}(\boldsymbol{x})) = E_{\boldsymbol{x}}(h_{\text{erm}}(\boldsymbol{x})) + \frac{1}{3}\sigma^2\gamma\|g_{\boldsymbol{x}}\|^2_{L_2(X)},$$
$$\overline{E}_{\boldsymbol{x}}(\mathcal{H}_{\widetilde{\text{PI}}}(\boldsymbol{x})) = E_{\boldsymbol{x}}(h_{\widetilde{\text{erm}}}(\boldsymbol{x})) + \frac{1}{3}\tilde{\sigma}^2\gamma\|\tilde{g}_{\boldsymbol{x}}\|^2_{L_2(\widetilde{X})}.$$

Averaging over $X \sim p_X$ via Lem. 5 gives $\mathbb{E}_X\left[\|g_X\|^2_{L_2(X)}\right] = k$ and $\mathbb{E}_X\left[\|\tilde{g}_X\|^2_{L_2(\widetilde{X})}\right] = \operatorname{tr}(\mathcal{S})$. Substituting into pointwise approx. errors and subtracting gives the desired result $f(1)$. □

H.3 PROOF OF THMS. 3.A AND 3.B—ROBUST BOUNDS WITH DA

Theorem 3 (robust bounds with DA)**.** *Under Asms. 1a–1b, suppose valid PI models following Asms. 2 and 3 yield respective intervals $\mathcal{H}_{\text{PI}}(\boldsymbol{x})$ and $\mathcal{H}_{\text{PI}+\text{IV}}(\boldsymbol{x})$ covering an $(\varepsilon, \mathcal{T})$-invariant $h_\star$. Then,*

(A) ε-padded post-DA intervals $\mathcal{H}_{\widetilde{\text{PI}}}(\boldsymbol{x}) \pm \varepsilon$ continue to remain valid for all $\varepsilon \geq 0$ on p_X-a.e. $\boldsymbol{x}$.

(B) ε-padded T-as-IV intervals $\mathcal{H}_{\widetilde{\text{PI}}+\widetilde{\text{IV}}}(\boldsymbol{x}) \pm \varepsilon$ are also valid for $\widetilde{Z} \in \{Z, (Z, T)\}$ on p_X-a.e. $\boldsymbol{x}$ if

$$\varepsilon \leq \tilde{\sigma}\sqrt{\gamma_{\boldsymbol{z}}} - \sigma\sqrt{\gamma^\star_{\boldsymbol{z}}}, \qquad \text{where } \gamma^\star_{\boldsymbol{z}} \leq \gamma_{\boldsymbol{z}} \text{ is the true IV budget.}$$

Proof. Sketch: (i) build a surrogate target $h_\dagger$ within ε of $h_\star$, (ii) certify $h_\dagger \in \mathcal{H}_{\widetilde{\text{PI}}}$ by DPI (Part A), and (iii) verify the IV moment via Minkowski (Part B); ε-pad to convert $h_\dagger$ membership into validity

Surrogate. Define the surrogate as the smoothed target $h_\dagger \coloneqq \mathbb{E}[h_\star(X) \,|\, \widetilde{X} = \cdot\,] \in \mathcal{H}$ (Asm. 1a). Define $\eta \coloneqq h_\star(X) - h_\star(\widetilde{X})$, with $|\eta| \leq \varepsilon$ a.s. by $(\varepsilon, \mathcal{T})$-invariance. Now, $h_\dagger$ obeys two identities:

$$\begin{aligned}
h_\dagger(\widetilde{X}) &= h_{\widetilde{\text{erm}}}(\widetilde{X}) - \mathbb{E}[U \,|\, \widetilde{X}], && \text{(Take } \mathbb{E}[\,\cdot\,|\widetilde{X}] \text{ in Eq. (1).)} \\
&= h_\star(\widetilde{X}) + \mathbb{E}[\eta \,|\, \widetilde{X}], && \text{(Take } \mathbb{E}[\,\cdot\,|\widetilde{X}] \text{ in def. of } \eta.) \\
\implies \big\|h_\dagger - h_\star\big\|_{L_\infty(X)} &\overset{\text{Asm. 1b}}{\leq} \big\|h_\dagger - h_\star\big\|_{L_\infty(\widetilde{X})} \overset{\text{Jensen}}{\leq} \varepsilon. && \text{(¶)}
\end{aligned}$$

Furthermore, by tower property $\mathbb{E}[h_\dagger(\widetilde{X})] = \mathbb{E}[h_\star(X)] = 0$, so surrogate is $\widetilde{X}$-centered $h_\dagger \in \mathcal{H}_{\widetilde{X}}$.

Part (A): $h_\dagger \in \mathcal{H}_{\widetilde{\text{PI}}}$. The surrogate's offset from the post-DA center is $h_{\widetilde{\text{erm}}} - h_\dagger = \mathbb{E}[U \,|\, \widetilde{X}]$, so

$$\begin{aligned}
\|h_{\widetilde{\text{erm}}} - h_\dagger\|^2_{L_2(\widetilde{X})} &= \operatorname{Var}\big(\mathbb{E}[U \,|\, \widetilde{X}]\big), && (\widetilde{X}\text{-centered } h_\dagger,\ h_{\widetilde{\text{erm}}},\ \Rightarrow \|\cdot\|^2_{L_2} = \operatorname{Var}(\cdot).) \\
&\leq \operatorname{Var}\big(\mathbb{E}[U \,|\, X]\big), && (\text{Contraction over Markov } U - X - \widetilde{X}.) \\
&\leq \gamma\sigma^2, && (\text{Baseline valid } h_\star \in \mathcal{H}_{\text{PI}} \text{ (Asm. 1a, 2).}) \\
&\leq \gamma\tilde{\sigma}^2. && (\text{DPI } \rho \geq 1.)
\end{aligned}$$

Post-DA, Lem. 2 gives $\mathcal{H}_{\widetilde{\text{PI}}}$ as the $\mathcal{H}_{\widetilde{X}} \cap L_2(\widetilde{X})$-ball of sq.-radius $\tilde{\sigma}^2\gamma$ about $h_{\widetilde{\text{erm}}}$. So $h_\dagger \in \mathcal{H}$ (Asm. 1a) gives $h_\dagger \in \mathcal{H}_{\widetilde{\text{PI}}} \Rightarrow h_\dagger(\boldsymbol{x}) \in \mathcal{H}_{\widetilde{\text{PI}}}(\boldsymbol{x})$. Hence $h_\star(\boldsymbol{x}) \in \mathcal{H}_{\widetilde{\text{PI}}}(\boldsymbol{x}) \pm \varepsilon$ for p_X-a.e. $\boldsymbol{x}$ by Eq. (¶).

Part (B): $h_\dagger \in \mathcal{H}_{\widetilde{\text{PI}}+\widetilde{\text{IV}}}$. To verify membership, we need to test the surrogate's residual $Y - h_\dagger(\widetilde{X})$ against the IV moment constraint in Asm. 3. Note that this residual is mean-zero and reads as

$$Y - h_\dagger(\widetilde{X}) = \zeta + U + \xi, \qquad \zeta \coloneqq \eta - \mathbb{E}[\eta \,|\, \widetilde{X}], \qquad \|\zeta\|^2_{L_2} = \mathbb{E}[\operatorname{Var}(\eta \,|\, \widetilde{X})] \leq \varepsilon^2.$$

For any choice of IV $\widetilde{Z} \in \{Z, (Z, T)\}$, the conditional $\mathbb{E}[U + \xi \,|\, Z] = \mathbb{E}[U + \xi \,|\, \widetilde{Z}]$ is conserved since $T \perp\!\!\!\perp (\xi, U, Z)$. Writing the true budget $\gamma^\star_{\boldsymbol{z}} \coloneqq \|\mathbb{E}[U + \xi \,|\, Z]\|^2_{L_2} \big/ \sigma^2 \leq \gamma_{\boldsymbol{z}}$ (Asm. 3), we have

$$\begin{aligned}
\Big\|\mathbb{E}[Y - h_\dagger(\widetilde{X}) \mid \widetilde{Z}]\Big\|_{L_2} &\leq \Big\|\mathbb{E}[\zeta \mid \widetilde{Z}]\Big\|_{L_2} + \Big\|\mathbb{E}[U + \xi \mid \widetilde{Z}]\Big\|_{L_2}, && \text{(Minkowski.)} \\
&\leq \varepsilon + \sigma\sqrt{\gamma^\star_{\boldsymbol{z}}}, && (\text{Contraction; } T \perp\!\!\!\perp (\xi, U, Z).) \\
&\leq \tilde{\sigma}\sqrt{\gamma_{\boldsymbol{z}}}. && (\varepsilon \leq \tilde{\sigma}\sqrt{\gamma_{\boldsymbol{z}}} - \sigma\sqrt{\gamma^\star_{\boldsymbol{z}}}.)
\end{aligned}$$

This is exactly the leaky-IV constraint of Asm. 3 under the augmented law, so together with Part (A), $h_\dagger \in \mathcal{H}_{\widetilde{\text{PI}}+\widetilde{\text{IV}}}$, and $h_\star(\boldsymbol{x}) \in \mathcal{H}_{\widetilde{\text{PI}}+\widetilde{\text{IV}}}(\boldsymbol{x}) \pm \varepsilon$ for p_X-a.e. $\boldsymbol{x}$ uniformly over $\widetilde{Z} \in \{Z, (Z, T)\}$. □

Remark 2 (conservativeness of Thm. 3.B.)**.** Minkowski assumes the worst-case where the two residual components are perfectly aligned. Instead, one can define

$$\kappa := \Big\langle \mathbb{E}[\zeta|\widetilde{Z}],\, \mathbb{E}[U{+}\xi|\widetilde{Z}] \Big\rangle_{L_2} \Big/ \Big(\big\|\mathbb{E}[\zeta|\widetilde{Z}]\big\| \cdot \big\|\mathbb{E}[U{+}\xi|\widetilde{Z}]\big\| \Big) \in [-1, 1],$$

as an alignment sensitivity parameter. The sufficient condition for post-DA leaky-IV validity is now

$$\varepsilon^2 + 2\kappa\,\varepsilon\,\sigma\sqrt{\gamma_z^\star} + \sigma^2\gamma_z^\star \;\le\; \tilde{\sigma}^2\gamma_z.$$

A conservative $\kappa = 1$ then recovers Thm. 3.B, while a plausible $\kappa = 0$ (misspecification uncorrelated to confounding) gives the Pythagorean budget $\varepsilon \le \sqrt{\tilde{\sigma}^2\gamma_z - \sigma^2\gamma_z^\star}$, strictly dominating the smaller $\tilde{\sigma}\sqrt{\gamma_z} - \sigma\sqrt{\gamma_z^\star}$. The observable diagnostic $\varepsilon \le (\tilde{\sigma} - \sigma)\sqrt{\gamma_z}$ of §D remains assumption-free.

H.4 PROOF OF PROP. 1—ESTIMATION WITH DA

Proposition 1 (estimation with DA)**.** *For $\mathcal{T}$-inv. $h_\star$, Asms. 1a–1c, with post-DA bias $\tilde{b} \coloneqq h_{\widetilde{\mathrm{erm}}} - h_\star$,*

$$E(h_{\mathrm{erm}}) - E(h_{\widetilde{\mathrm{erm}}}) \quad = \quad \underbrace{\sigma^2 \mathcal{S}[\tilde{b}] \cdot (\rho - 1)}_{\textit{gain}} + E(h_{\mathrm{erm}}) \cdot \Big(\underbrace{1 - \mathcal{K}[\tilde{b}]}_{\textit{lever}} - \underbrace{(\delta^* \delta)[\tilde{b}]}_{\textit{penalty}} \Big).$$

Proof. Under Asm. 1a, $h_{\mathrm{erm}} = \mathbb{E}[Y|X]$ and $h_{\widetilde{\mathrm{erm}}} = \mathbb{E}[Y|\widetilde{X}]$. Taking conditional means in Eq. (1),

$$b = \mathbb{E}[Y|X] - h_\star = \mathbb{E}[U|X], \qquad \tilde{b} = \mathbb{E}[Y|\widetilde{X}] - h_\star = \mathbb{E}[U|\widetilde{X}],$$

where the latter follows from $\mathcal{T}$-inv. of $h_\star$. Note that $b, \tilde{b} \in \mathcal{H}$ since $h_{\mathrm{erm}}, h_{\widetilde{\mathrm{erm}}}, h_\star \in \mathcal{H}$ (Asm. 1a). Further, $\mathbb{E}[U] = 0$ implies $b \in \mathcal{H}_X$ and $\tilde{b} \in \mathcal{H}_{\widetilde{X}}$. Now, re-write the error terms in respective biases:

$$E(h_{\mathrm{erm}}) = \|b\|^2_{L_2(X)}, \qquad E(h_{\widetilde{\mathrm{erm}}}) = \|\tilde{b}\|^2_{L_2(X)}.$$

On $\widetilde{X}$-centered functions $h \in L^0_2(\widetilde{X})$, define the shift ratio $s(h) \coloneqq \|h\|^2_{L_2(X)} \big/ \|h\|^2_{\widetilde{X}}$ ($s(0) \coloneqq 0$). Substitute into $E(h_{\widetilde{\mathrm{erm}}})$:

$$\begin{aligned}
E(h_{\widetilde{\mathrm{erm}}}) &= \|\tilde{b}\|^2_{L_2(X)}, \\
&= s(\tilde{b}) \cdot \|\tilde{b}\|^2_{\widetilde{X}}, && \text{(Multiply and divide by } \|\tilde{b}\|^2_{\widetilde{X}}.) \\
&= s(\tilde{b}) \cdot \|\tilde{b}\|^2_{L_2(\widetilde{X})}, && (\tilde{b} \in \mathcal{H}_{\widetilde{X}} \implies \|\tilde{b}\|^2_{L_2(\widetilde{X})} = \|\tilde{b}\|^2_{\widetilde{X}}.) \\
&= s(\tilde{b}) \cdot \Big(\|b\|^2_{L_2(X)} - (\tilde{\sigma}^2 - \sigma^2) \Big), && \text{(Prop. 4.)} \\
&= s(\tilde{b}) \cdot \big(E(h_{\mathrm{erm}}) - \sigma^2 \cdot (\rho - 1) \big), && \text{(Substitute } E(h_{\mathrm{erm}}) = \|b\|^2_{L_2(X)} \text{ and } \rho = \tilde{\sigma}^2/\sigma^2.)
\end{aligned}$$

Finally, subtract this from $E(h_{\mathrm{erm}})$ and re-arrange to get a general form of the proposition statement:

$$E(h_{\mathrm{erm}}) - E(h_{\widetilde{\mathrm{erm}}}) = \sigma^2 s(\tilde{b}) \cdot (\rho - 1) + E(h_{\mathrm{erm}}) \cdot (1 - s(\tilde{b})).$$

Specializing under Asms. 1b–1c. Recall that Asms. 1b–1c gives $\|h\|^2_{L_2(X)} = \langle h, \mathcal{S}h \rangle_{\widetilde{X}}$ where operator $\mathcal{S}$ has a Rayleigh quotient $\mathcal{S}[h] = \langle h, \mathcal{S}h \rangle_{\widetilde{X}} \big/ \|h\|^2_{\widetilde{X}}$ and decomposition $\mathcal{S} = \mathcal{K} + \delta^* \delta$ by Lem. 1.

Trivially, substituting $s(\tilde{b}) = \mathcal{S}[\tilde{b}]$ into the above general identity gives the result, with $\mathcal{S}[\tilde{b}] = \mathcal{K}[\tilde{b}] + (\delta^* \delta)[\tilde{b}]$ (Lem. 1 and Rayleigh quotient is linear in the operator). □

H.5 PROOF OF PROP. 2—SHARPER BOUNDS WITH DA

Proposition 2 (sharper bounds with DA). *Denoting $L_2(X)$-volume by $|\cdot|$, under Asms. 1b–1c and 2,*

$$\frac{|\mathcal{H}_{\widetilde{\text{PI}}}|}{|\mathcal{H}_{\text{PI}}|} \le \Big(\rho \cdot \operatorname{tr}(\mathcal{S})/\,k\Big)^{k/2}, \qquad \frac{\mathbb{E}\big[|\mathcal{H}_{\widetilde{\text{PI}}}(X)|^2\big]}{\mathbb{E}\big[|\mathcal{H}_{\text{PI}}(X)|^2\big]} = \underbrace{\rho}_{\textit{price}} \cdot \operatorname{tr}\Big(\underbrace{\mathcal{K}}_{\textit{lever}} + \underbrace{\delta^*\delta}_{\textit{penalty}} \Big) \Big/ k.$$

Proof. Sketching our steps: (i) **volume:** write the post-DA ball in an $L_2(X)$-orthonormal basis of $\mathcal{H}_{\widetilde{X}}$ and compare via AM-GM; (ii) **width:** write widths as representer norms and average via Lem. 5.

Setup. Write Lem. 2 and Cor. 3 pre- and post-DA identified sets in their respective L_2 geometries,

$$\mathcal{H}_{\text{PI}} = \Big\{h \in \mathcal{H}_X : \|h - h_{\text{erm}}\|^2_{L_2(X)} \le \sigma^2\gamma\Big\}, \qquad \mathcal{H}_{\text{PI}}(\boldsymbol{x}) = \{h_{\text{erm}}(\boldsymbol{x})\} \pm \sigma\sqrt{\gamma}\|g_{\boldsymbol{x}}\|_{L_2(X)},$$
$$\mathcal{H}_{\widetilde{\text{PI}}} = \Big\{h \in \mathcal{H}_{\widetilde{X}} : \|h - h_{\widetilde{\text{erm}}}\|^2_{L_2(\widetilde{X})} \le \tilde{\sigma}^2\gamma\Big\}, \qquad \mathcal{H}_{\widetilde{\text{PI}}}(\boldsymbol{x}) = \{h_{\widetilde{\text{erm}}}(\boldsymbol{x})\} \pm \tilde{\sigma}\sqrt{\gamma}\|\tilde{g}_{\boldsymbol{x}}\|_{L_2(\widetilde{X})}.$$

We measure both sets in ambient $L_2(X)$-volume and -width by bridging with $L_2(\widetilde{X})$ using operator $\mathcal{S}$.

Volume. Define the $L_2(X)$-volume of a set inside a k-dim. subspace as the Lebesgue volume of its coordinates in *any* $L_2(X)$-orthonormal basis of that subspace; in particular $V_k \coloneqq \pi^{k/2}/\,\Gamma(k/2+1)$, the Euclidean unit-ball volume, is the $L_2(X)$-volume of a k-dim. unit ball in any $L_2(X)$-orthonormal basis. The baseline set is an $L_2(X)$-round ball of radius $\sigma\sqrt{\gamma}$ within $\mathcal{H}_X$, so directly

$$|\mathcal{H}_{\text{PI}}| = V_k(\sigma\sqrt{\gamma})^k.$$

For the post-DA set, take an $L_2(\widetilde{X})$-orthonormal eigenbasis $\{\psi_i\}_{i=1}^k$ of $\mathcal{S}$ on $\mathcal{H}_{\widetilde{X}}$ with eigenvalues $\lambda_i > 0$ (Asms. 1b–1c), and rescale $e_i \coloneqq \psi_i/\sqrt{\lambda_i}$. First, notice $\{e_i\} \subset \mathcal{H}_{\widetilde{X}}$ is $L_2(X)$-orthonormal:

$$\begin{aligned}\langle e_i, e_j\rangle_{L_2(X)} &= \langle e_i, \mathcal{S}e_j\rangle_{L_2(\widetilde{X})}, && \text{(Def. of } \mathcal{S}.)\\ &= \frac{\lambda_j}{\sqrt{\lambda_i\lambda_j}}\langle \psi_i, \psi_j\rangle_{L_2(\widetilde{X})} = \mathrm{Id}_{ij}. && \text{(Def. of } e_i, \text{ and } \mathcal{S}\psi_j = \lambda_j\psi_j.)\end{aligned}$$

Therefore we can measure $L_2(X)$-volume in the coordinates $h - h_{\widetilde{\text{erm}}} = \sum_i v_i e_i$. Second, we have:

$$\langle e_i, e_j\rangle_{L_2(\widetilde{X})} = \frac{1}{\sqrt{\lambda_i\lambda_j}}\langle \psi_i, \psi_j\rangle_{L_2(\widetilde{X})} = \frac{1}{\lambda_i}\,\mathrm{Id}_{ij}. \qquad (L_2(\widetilde{X})\text{-orthonormality of } \psi_i.)$$

Using this identity and linearity of inner-product, substitute $h - h_{\widetilde{\text{erm}}} = \sum_i v_i e_i$ into the def. of $\mathcal{H}_{\widetilde{\text{PI}}}$:

$$\|h - h_{\widetilde{\text{erm}}}\|^2_{L_2(\widetilde{X})} \;=\; \left\langle \sum_i v_i e_i, \sum_j v_j e_j \right\rangle_{L_2(\widetilde{X})} = \sum_i\sum_j v_i v_j \langle e_i, e_j\rangle_{L_2(\widetilde{X})} \;=\; \boxed{\sum_{i=1}^k \frac{v_i^2}{\lambda_i} \le \tilde{\sigma}^2\gamma.}$$

Out pops an ellipsoid with semi-axes $\tilde{\sigma}\sqrt{\gamma}\cdot\sqrt{\lambda_i}$. Now, taking its volume and subsequently the ratio:

$$|\mathcal{H}_{\widetilde{\text{PI}}}| \;=\; V_k(\tilde{\sigma}\sqrt{\gamma})^k \prod_i \sqrt{\lambda_i} \;=\; V_k(\tilde{\sigma}\sqrt{\gamma})^k (\det\mathcal{S})^{1/2},$$

$$\implies \frac{|\mathcal{H}_{\widetilde{\text{PI}}}|}{|\mathcal{H}_{\text{PI}}|} \;=\; \rho^{\frac{k}{2}}\,(\det\mathcal{S})^{\frac{1}{2}} \;=\; \rho^{\frac{k}{2}}\Big(\prod_i \lambda_i\Big)^{\frac{1}{2}} \overset{\text{AM–GM}}{\le} \rho^{\frac{k}{2}}\Big(\frac{1}{k}\sum_i \lambda_i\Big)^{\frac{k}{2}} \;=\; \Big(\rho\cdot\frac{\operatorname{tr}(\mathcal{S})}{k}\Big)^{\frac{k}{2}},$$

with equality iff eigenvalues λ_i coincide, i.e. iff $\mathcal{S} = \lambda\cdot\mathrm{Id}$ for some $\lambda > 0$.

Width. Cor. 3 gives half-widths at a query $\boldsymbol{x}$ as $\sigma\sqrt{\gamma}\|g_{\boldsymbol{x}}\|_{L_2(X)}$ and $\tilde{\sigma}\sqrt{\gamma}\|\tilde{g}_{\boldsymbol{x}}\|_{L_2(\widetilde{X})}$. Averaging their squares over $X \sim p_X$ using Lem. 5,

$$\mathbb{E}_X\Big[\,|\mathcal{H}_{\text{PI}}(X)|^2\,\Big] = 4\sigma^2\gamma\,\mathbb{E}_X\Big[\,\|g_X\|^2_{L_2(X)}\,\Big] = 4\sigma^2\gamma\,k,$$
$$\mathbb{E}_X\Big[\,|\mathcal{H}_{\widetilde{\text{PI}}}(X)|^2\,\Big] = 4\tilde{\sigma}^2\gamma\,\mathbb{E}_X\Big[\,\|\tilde{g}_X\|^2_{L_2(\widetilde{X})}\,\Big] = 4\tilde{\sigma}^2\gamma\,\operatorname{tr}(\mathcal{S}).$$

Splitting $\operatorname{tr}(\mathcal{S}) = \operatorname{tr}(\mathcal{K}) + \operatorname{tr}(\delta^*\delta)$ (Lem. 1), their ratio gives the claimed identity. □

H.6 PROOF OF PROP. 3—SHAPE CONSTRAINT VS. CHANGE-OF-MEASURE

Proposition 3 (shape constraint vs. change-of-measure)**.** *Under Asms. 1a–1b, and Asms. 2 to 5 with $\mathcal{T}$-invariant $h_\star$ and $Z = \varnothing$. Set $\varepsilon = \gamma_{\boldsymbol{z}} = 0$, and suppose the baseline PI set is valid $h_\star \in \mathcal{H}_{\text{pi}}$. Then, $h_{\widetilde{\text{erm}}} = \mathcal{T}h_{\text{erm}} \in \mathcal{H}_{\text{inv}}$, where $\mathcal{T}$ is the $L_2(X)$-orthogonal projection onto $\mathcal{I}$ (cf. §H). Further,*

$$\mathcal{H}_{\widetilde{\text{pi}}} = \Big\{h \in \mathcal{H}_{\widetilde{X}} \;:\; \big\|h - h_{\widetilde{\text{erm}}}\big\|_{\widetilde{X}}^2 \le \tilde{\sigma}^2\gamma\Big\},$$
$$\mathcal{H}_{\widetilde{\text{pi}}+\widetilde{\text{iv}}} = \Big\{h \in \mathcal{H}_{\text{inv}} \colon \big\|h - h_{\widetilde{\text{erm}}}\big\|_{\widetilde{X}}^2 \le \tilde{\sigma}^2\gamma\Big\},$$
$$\mathcal{H}_{\text{pi}+\text{inv}} = \Big\{h \in \mathcal{H}_{\text{inv}} \colon \big\|h - h_{\widetilde{\text{erm}}}\big\|_{X}^2 \le \sigma^2(1+\gamma-\rho)\Big\}.$$

All identified sets are balls centered about $h_{\widetilde{\text{erm}}}$. Both $\mathcal{H}_{\widetilde{\text{pi}}+\widetilde{\text{iv}}}$ and $\mathcal{H}_{\text{pi}+\text{inv}}$ live on the slice $\mathcal{H}_{\text{inv}}$, the latter with a smaller radius $\sigma\sqrt{1+\gamma-\rho} \le \tilde{\sigma}\sqrt{\gamma}$ with equality iff $\rho = 1$. It trivially follows that,

$$h_\star \in \mathcal{H}_{\text{pi}+\text{inv}} \subseteq \mathcal{H}_{\widetilde{\text{pi}}+\widetilde{\text{iv}}} \subseteq \mathcal{H}_{\widetilde{\text{pi}}},$$

and the same dominance hierarchy holds for sharpness and worst-case errors over $\mathcal{H}_\bullet$ and $\mathcal{H}_\bullet(\boldsymbol{x})$.

Proof. Fix the common center and geometry, characterize the three sets as balls, then compare radii:

Shared center. Under Asms. 1a–1b and 4, Lem. 4 gives: $h_{\widetilde{\text{erm}}} = \mathcal{T}h_{\text{erm}} \in \mathcal{H}_{\text{inv}}$.

The post-DA set. $\mathcal{H}_{\widetilde{\text{pi}}} = \Big\{h \in \mathcal{H}_{\widetilde{X}} \colon \|h - h_{\widetilde{\text{erm}}}\|_{\widetilde{X}}^2 \le \tilde{\sigma}^2\gamma\Big\}$ immediate from Lem. 2 under $p_{\widetilde{X}Y}$.

The T-as-IV set. For $\widetilde{Z} = T$, $\gamma_{\boldsymbol{z}} = 0$, the IV constraint of Cor. 2 under $p_{\widetilde{X}Y\widetilde{Z}}$ filters $h \in \mathcal{H}_{\widetilde{X}}$ via

$$\begin{aligned}
&\operatorname{Var}(\mathbb{E}[Y - h(\widetilde{X})\,|\,T]) = 0 \iff \operatorname{Var}(\mathbb{E}[Y] - \mathbb{E}[h(\widetilde{X})\,|\,T]) = 0 && (T \perp\!\!\!\perp Y \text{ by construction.})\\
\iff\quad &\operatorname{Var}(\mathbb{E}[h(\widetilde{X})\,|\,T]) = 0 \iff \mathcal{E}_{\widetilde{X}|T}h = 0 && (\text{Def. of } \mathcal{E}_{\widetilde{X}|T}.)\\
&\iff h \in \operatorname{null}\big(\mathcal{E}_{\widetilde{X}|T}\,\big|_{\mathcal{H}_{\widetilde{X}}}\big) = \mathcal{H}_{\text{inv}}, && (\text{Asm. 5.})\\
&\implies \mathcal{H}_{\widetilde{\text{pi}}+\widetilde{\text{iv}}} = \mathcal{H}_{\widetilde{\text{pi}}} \cap \mathcal{H}_{\text{inv}}. && (\text{Cor. 2 under } p_{\widetilde{X}Y\widetilde{Z}}.)
\end{aligned}$$

The inv. constraint set. By Cor. 1, $\mathcal{H}_{\text{pi}+\text{inv}} = \mathcal{H}_{\text{pi}} \cap \mathcal{H}_{\text{inv}}$, with $\mathcal{H}_{\text{pi}}$ the X-ball of radius $\sigma\sqrt{\gamma}$ about h_{erm} (Lem. 2). For invariant $h \in \mathcal{H}_{\text{inv}}$, using Pythagoras along $\mathcal{T}$ followed by $h = \mathcal{T}h$,

$$\begin{aligned}
\|h - h_{\text{erm}}\|_X^2 &= \|\mathcal{T}(h - h_{\text{erm}})\|_X^2 + \|(\mathrm{Id} - \mathcal{T})(h - h_{\text{erm}})\|_X^2,\\
&= \|h - \mathcal{T}h_{\text{erm}}\|_X^2 + \|(\mathrm{Id} - \mathcal{T})h_{\text{erm}}\|_X^2. && (\triangle)
\end{aligned}$$

Writing pre- and post-DA biases $b \coloneqq h_{\text{erm}} - h_\star$ and $\tilde{b} \coloneqq h_{\widetilde{\text{erm}}} - h_\star$, the second term reduces to:

$$\begin{aligned}
\|(\mathrm{Id} - \mathcal{T})h_{\text{erm}}\|_X^2 &= \|(\mathrm{Id} - \mathcal{T})(h_\star + b)\|_X^2 = \|(\mathrm{Id} - \mathcal{T})b\|_X^2, && (\text{Def. of bias } b, \mathcal{T}\text{-inv. } h_\star.)\\
&= \|b - \tilde{b}\|_X^2 = \|b\|_X^2 - \|\tilde{b}\|_X^2, && (\text{Lem. 4 } \tilde{b} = \mathcal{T}b, \text{Pythagoras.})\\
&= \|b\|_X^2 - \|\tilde{b}\|_{\widetilde{X}}^2 = \tilde{\sigma}^2 - \sigma^2. && (\tilde{b} \in \mathcal{I} \Rightarrow \text{Eq. (6), Prop. 4.})
\end{aligned}$$

Substituting into ($\triangle$), together with $\mathcal{T}h_{\text{erm}} = h_{\widetilde{\text{erm}}}$ and baseline ball constraint $\|h - h_{\text{erm}}\|_X^2 \le \sigma^2\gamma$,

$$\implies \|h - h_{\widetilde{\text{erm}}}\|_X^2 \;\le\; \sigma^2\gamma - (\tilde{\sigma}^2 - \sigma^2) \;=\; \sigma^2(1+\gamma-\rho).$$

Inclusion and dominance hierarchy. Both T-as-IV and inv.-constrained sets $\mathcal{H}_{\widetilde{\text{pi}}+\widetilde{\text{iv}}}$, $\mathcal{H}_{\text{pi}+\text{inv}}$ are balls in $\mathcal{H}_{\text{inv}}$ about the common center $h_{\widetilde{\text{erm}}}$. Since $h - h_{\widetilde{\text{erm}}} \in \mathcal{I}$ for invariant h, their norms are measured in the common geometry of Eq. (6), with squared radii obeying

$$\tilde{\sigma}^2\gamma - \sigma^2(1+\gamma-\rho) = \sigma^2(1+\gamma)(\rho-1) \;\ge\; 0,$$

where $\rho \ge 1$ by the DPI (Prop. 4). Hence $\mathcal{H}_{\text{pi}+\text{inv}} \subseteq \mathcal{H}_{\widetilde{\text{pi}}+\widetilde{\text{iv}}}$, while $\mathcal{H}_{\widetilde{\text{pi}}+\widetilde{\text{iv}}} \subseteq \mathcal{H}_{\widetilde{\text{pi}}}$ is immediate.

Finally, set inclusion directly implies a dominance hierarchy in sharpness and worst-case errors. □

H.7 PROOF OF PROP. 4—SOFT INTERVENTION WITH DA

Proposition 4 (soft intervention with DA)**.** *For $\mathcal{T}$-inv. $h_\star$, DA acts on Eq. (1) as a* soft-intervention*:*

(A) Structural equation: *Post-DA variables $(\xi, U, \widetilde{X}, Y, \widetilde{Z})$ with $\widetilde{Z} \in \{Z, (Z,T)\}$ follow Eq. (5):*

$$Y = h_\star(\widetilde{X}) + U + \xi, \qquad \mathbb{E}[U] = \mathbb{E}[\xi] = 0, \qquad \xi \perp\!\!\!\perp (U, \widetilde{X}, \widetilde{Z}).$$

(B) Markov transfer: *Let p be Markov w.r.t. a DAG $\mathcal{G}$ over $V \coloneqq \{U, X, Y, Z_1, Z_2, \ldots\}$ where $\mathrm{NonDesc}_{\mathcal{G}}(X) = V \backslash \{X, Y\}$. Then, $p_{\widetilde{V}}$ is Markov w.r.t. $\widetilde{\mathcal{G}}$, where $\widetilde{V}, \widetilde{\mathcal{G}}$ relabel X to $\widetilde{X}$ in $V, \mathcal{G}$.*

(C) Attenuated confounding: *Under Asm. 1a, let biases $b \coloneqq h_{\mathrm{erm}} - h_\star$ and $\tilde{b} \coloneqq h_{\widetilde{\mathrm{erm}}} - h_\star$. Then,*

$$\|b\|^2_{L_2(X)} - \|\tilde{b}\|^2_{L_2(\widetilde{X})} \;=\; \tilde{\sigma}^2 - \sigma^2 \;\geq\; 0.$$

Proof. Part *(a)* is immediate from def. of $\mathcal{T}$-inv. and $T \perp\!\!\!\perp (\xi, U, X, Y, Z)$. We prove parts *(b)–(c)*.

Part (b): Markov transfer. Write $W \coloneqq V \backslash \{X, Y\}$, so $\widetilde{V} = W \cup \{\widetilde{X}, Y\}$. Now, disintegrate $p_{\widetilde{V}}$,

$$p_{W\widetilde{X}Y} = p_W \otimes p_{\widetilde{X}|W} \otimes p_{Y|\widetilde{X}W},$$

and reduce each kernel to its $\widetilde{\mathcal{G}}$-kernel,

$$\begin{aligned}
&p_W = \bigotimes\nolimits_{V \in W} p_{V|\mathrm{Pa}_{\mathcal{G}}(V)}, && (W \text{ ancestral in } \mathcal{G}, \text{ so its } \mathcal{G}\text{-marginal factorizes.})\\
&p_{\widetilde{X}|W}(\,\cdot\,|\,\boldsymbol{w}) && (\text{Integrate out } \boldsymbol{\tau}.)\\
&= \int \alpha(\cdot, \boldsymbol{\tau})_{\#}\, p_{X|W}(\,\cdot\,|\,\boldsymbol{w})\; p_T(\mathrm{d}\boldsymbol{\tau}), && (\text{Use } T \perp\!\!\!\perp (W, X) \text{ and } \widetilde{X} = \alpha(X, \boldsymbol{\tau}) \text{ on } \{T = \boldsymbol{\tau}\}.)\\
&= \int \alpha(\cdot, \boldsymbol{\tau})_{\#}\, p_{X|\mathrm{Pa}_{\mathcal{G}}(X)}\big(\,\cdot\,\big|\,\mathrm{pa}_{\mathcal{G}}(\boldsymbol{x})\big)\; p_T(\mathrm{d}\boldsymbol{\tau}), && (W \text{ non-desc., so } X \perp\!\!\!\perp W \backslash \mathrm{Pa}_{\mathcal{G}}(X) \,\big|\, \mathrm{Pa}_{\mathcal{G}}(X).)\\
&=: p_{\widetilde{X}|\mathrm{Pa}_{\mathcal{G}}(X)}\big(\,\cdot\,\big|\,\mathrm{pa}_{\mathcal{G}}(\boldsymbol{x})\big),\\
&p_{Y|\widetilde{X}W}(\,\cdot\,|\,\boldsymbol{x}, \boldsymbol{w}) && (\text{Part } (a)\text{: } Y = h_\star(\widetilde{X}) + U + \xi \text{ with exogenous } \xi \perp\!\!\!\perp (W, \widetilde{X}).)\\
&= p_\xi(\,\cdot\, - h_\star(\boldsymbol{x}) - u) =: p_{Y|\widetilde{X}U}(\,\cdot\,|\,\boldsymbol{x}, u).
\end{aligned}$$

Substitute back to get the desired factorization along $\widetilde{\mathcal{G}}$, i.e. all parent sets of $\mathcal{G}$ with X relabeled $\widetilde{X}$:

$$p_{W\widetilde{X}Y} = \bigotimes\nolimits_{V \in W} p_{V|\mathrm{Pa}_{\mathcal{G}}(V)} \;\otimes\; p_{\widetilde{X}|\mathrm{Pa}_{\mathcal{G}}(X)} \;\otimes\; p_{Y|\widetilde{X}U}.$$

Part (c): attenuated confounding. By Asm. 1a, $h_{\mathrm{erm}} = \mathbb{E}[Y|X]$, $h_{\widetilde{\mathrm{erm}}} = \mathbb{E}[Y|\widetilde{X}]$. Taking conditional means in Eq. (1),

$$b = \mathbb{E}[Y|X] - h_\star = \mathbb{E}[U|X], \qquad \tilde{b} = \mathbb{E}[Y|\widetilde{X}] - h_\star = \mathbb{E}[U|\widetilde{X}],$$

where the latter follows from $\mathcal{T}$-inv. of $h_\star$.

The identity: Since $\xi \perp\!\!\!\perp (U, X)$, the noise floor σ^2 decomposes as

$$\begin{aligned}
\sigma^2 &= \mathbb{E}[\mathrm{Var}(Y|X)], && (\text{Asm. 1a.})\\
&= \mathrm{Var}(\xi) + \mathbb{E}[\mathrm{Var}(U|X)], && (\xi \perp\!\!\!\perp (U, X).)\\
&= \mathrm{Var}(\xi) + \mathrm{Var}(U) - \|b\|^2_{L_2(X)}, && (\text{Law of total variance.})
\end{aligned}$$

and identical steps under $\widetilde{X}$ ($\mathcal{T}$-inv. at the first step) gives $\tilde{\sigma}^2 = \mathrm{Var}(\xi) + \mathrm{Var}(U) - \|\tilde{b}\|^2_{L_2(\widetilde{X})}$. Subtracting the two noise floors gives $\|b\|^2_{L_2(X)} - \|\tilde{b}\|^2_{L_2(\widetilde{X})} = \tilde{\sigma}^2 - \sigma^2$.

Non-negativity: As $\widetilde{X} = \alpha(X, T)$ with $T \perp\!\!\!\perp (U, X)$, the chain $U - X - \widetilde{X}$ is Markov, hence:

$$\tilde{b} = \mathbb{E}[U \,|\, \widetilde{X}] = \mathbb{E}\Big[\mathbb{E}[U|X]\,\Big|\,\widetilde{X}\Big] = \mathbb{E}[b(X) \,|\, \widetilde{X}] \qquad (\text{Tower rule.})$$

Because conditional expectation is an L_2-contraction, we have:

$$\|\tilde{b}\|^2_{L_2(\widetilde{X})} = \mathrm{Var}(\mathbb{E}[U \,|\, \widetilde{X}]) \leq \mathrm{Var}(\mathbb{E}[U \,|\, X]) = \|b\|^2_{L_2(X)},$$

i.e. $\tilde{\sigma}^2 - \sigma^2 = \|b\|^2_{L_2(X)} - \|\tilde{b}\|^2_{L_2(\widetilde{X})} \geq 0$. □

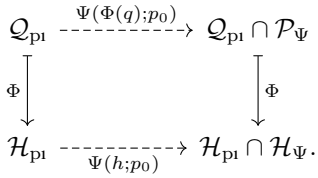


Figure 7: A picture of Prop. 5: Filtering baseline models $q \in \mathcal{Q}_{\text{PI}}$ by the predicate $\Psi(\Phi(q); p_0)$ and then mapping through Φ yields the same hypothesis set as mapping $\mathcal{H}_{\text{PI}} = \Phi(\mathcal{Q}_{\text{PI}})$ first and then filtering by the corresponding function-level predicate $\Psi(h; p_0)$. Dashed arrows denote set restriction.

H.8 PROOF OF PROP. 5—SHARP BOUNDS WITH FIBERWISE CONSTRAINTS

Proposition 5 (sharp bounds with fiberwise constraints)**.** *Let p_0 be some fixed distribution and define*

$$\Phi\colon \mathcal{P}_{\text{PI}} \to \mathcal{H},\ q \mapsto h^q_\star.$$

For a set of baseline PI candidate distributions $\mathcal{Q}_{\text{PI}} \subseteq \mathcal{P}_{\text{PI}}$, their candidate hypotheses are given by

$$\mathcal{H}_{\text{PI}} := \Phi(\mathcal{Q}_{\text{PI}}).$$

Let $\Psi(h; p_0)$ be a predicate depending only on functions $h \in \mathcal{H}$ and the fixed law p_0. Define the sets

$$\mathcal{H}_\Psi := \{h \in \mathcal{H}\colon \Psi(h; p_0)\}, \qquad \mathcal{P}_\Psi := \{q \in \mathcal{P}_{\text{PI}}\colon \Psi(\Phi(q); p_0)\}.$$

Then $\mathcal{P}_\Psi = \Phi^{-1}(\mathcal{H}_\Psi)$ is a union of fibers of Φ above $\mathcal{H}_\Psi$, and the constrained identified set is sharp*:*

$$\Phi(\mathcal{Q}_{\text{PI}} \cap \mathcal{P}_\Psi) = \mathcal{H}_{\text{PI}} \cap \mathcal{H}_\Psi.$$

The same identity holds for any finite collection of such predicates. Fig. 7 states the prop. in a picture.

Proof. Re-write the claimed identity by substituting $\mathcal{H}_{\text{PI}} = \Phi(\cdot)$, $\mathcal{P}_\Psi = \Phi^{-1}(\cdot)$, $Q = \mathcal{Q}_{\text{PI}}$, $H = \mathcal{H}_\Psi$,

$$\Phi\big(Q \cap \Phi^{-1}(H)\big) = \Phi(Q) \cap H.$$

Necessity ($\subseteq$). Trivially, for any $h \in \Phi\big(Q \cap \Phi^{-1}(H)\big)$, we have $h = \Phi(q)$ for some candidate $q \in Q \cap \Phi^{-1}(H)$. Directly implying that $h \in \Phi(Q)$ and $h \in H$. Hence, the LHS implies the RHS.

Sufficiency ($\supseteq$). Let $h \in \Phi(Q) \cap H$. Then there exists $q \in Q$ such that $\Phi(q) = h$, which is in H. Since $\Phi(q) \in H$, then $q \in \Phi^{-1}(H)$. Thus, $q \in Q \cap \Phi^{-1}(H)$, giving $h = \Phi(q) \in \Phi\big(Q \cap \Phi^{-1}(H)\big)$.

□

Corollary 1 (sharp bounds with inv. constraints)**.** *Define the ε-approximate $\mathcal{T}$-invariance hypotheses*

$$\mathcal{H}^\varepsilon_{\text{INV}} := \big\{h \in \mathcal{H} : E_{\text{INV}}(h) \le \varepsilon^2\big\}.$$

Then, the explicit invariance constraint PI model from §4 is sharp*, meaning $\mathcal{H}_{\text{PI}+\text{INV}} = \mathcal{H}_{\text{PI}} \cap \mathcal{H}^\varepsilon_{\text{INV}}$.*

Proof. Immediate from Prop. 5 using $p_0 = p_{X\tilde{X}}$ and $\Psi(h; p_0) = \mathbf{1}\big\{E_{\text{INV}}(h) \le \varepsilon^2\big\}$. □

Corollary 2 (sharp bounds with IV constraints)**.** *Let σ_p^2 be the ERM noise floor under true p. Define*

$$\mathcal{H}_{\text{IV}} := \Big\{h \in \mathcal{H} : \text{Var}_p\Big(\mathbb{E}_p[Y - h(X) \,|\, Z]\Big) \le \sigma_p^2 \gamma_{\boldsymbol{z}}\Big\},$$

as the leaky-IV hypotheses. Then, the PI model of Asm. 3 is sharp*, meaning $\mathcal{H}_{\text{PI}+\text{IV}} = \mathcal{H}_{\text{PI}} \cap \mathcal{H}_{\text{IV}}$.*

Proof. Immediate from Prop. 5 using $p_0 = p_{XYZ}$ and the predicate

$$\Psi(h; p_0) = \mathbf{1}\Big\{\text{Var}_{p_0}\Big(\mathbb{E}_{p_0}[Y - h(X) \,|\, Z]\Big) \le \sigma_{p_0}^2 \gamma_{\boldsymbol{z}}\Big\}.$$

□

H.9 PROOF OF LEM. 1—SHIFT DECOMPOSITION

Lemma 1 (shift decomposition). *Under Asms. 1b–1c, $\mathcal{S} = \mathcal{K} + \delta^*\delta$, with $\mathcal{K} \succ 0$, $\delta^*\delta \succcurlyeq 0$ s.t. $\operatorname{rank}(\delta^*\delta) \le 1$, and $\operatorname{tr}(\delta^*\delta) = D^2_{\mathcal{B}}(p_X, p_{\widetilde{X}})$ is IPM^2 over the unit ball $\mathcal{B} := \{h \in \mathcal{H}_{\widetilde{X}} : \|h\|_{\widetilde{X}} \le 1\}$.*

Proof. We make three moves: (i) decompose the evaluation form as covariance-plus-mean, (ii) read off the two operators it splits into via Riesz, and (iii) identify the rank and trace of the mean part.

Evaluation form identity. Take $f, g \in \mathcal{H}_{\widetilde{X}}$. Scoring them in $L_2(X)$ gives

$$\begin{aligned}\langle f, g\rangle_{L_2(X)} &= \mathbb{E}\big[f(X)\, g(X)\big],\\ &= \operatorname{Cov}(f(X), g(X)) + \mathbb{E}[f(X)]\,\mathbb{E}[g(X)],\\ &= \langle f, g\rangle_X + (\delta f)\,(\delta g).\end{aligned}$$

Operator decomposition via Riesz. By Asm. 1c, $(\mathcal{H}_X, \langle\cdot,\cdot\rangle_X)$ is Hilbert. By Asm. 1b (cf. §H),

$$(\mathcal{H}_{\widetilde{X}}, \langle\cdot,\cdot\rangle_{\widetilde{X}})$$

is also Hilbert, so every symmetric bilinear form on it is represented by a unique self-adjoint operator.

Additionally, the forms $\langle\cdot,\cdot\rangle_{L_2(X)}$ and $\langle\cdot,\cdot\rangle_X$ are also both positive-definite on $\mathcal{H}_{\widetilde{X}}$ (§H).

Riesz gives unique self-adjoint, positive-definite $\mathcal{S}, \mathcal{K} : \mathcal{H}_{\widetilde{X}} \to \mathcal{H}_{\widetilde{X}}$ satisfying

$$\langle f, \mathcal{S}g\rangle_{\widetilde{X}} = \langle f, g\rangle_{L_2(X)}, \qquad\qquad \langle f, \mathcal{K}g\rangle_{\widetilde{X}} = \langle f, g\rangle_X.$$

Likewise, $\delta : \mathcal{H}_{\widetilde{X}} \to \mathbb{R}$ being a bounded linear functional has a unique representer $\delta_\sharp \in \mathcal{H}_{\widetilde{X}}$ s.t.

$$\delta h = \langle h, \delta_\sharp\rangle_{\widetilde{X}}, \qquad \forall\, h \in \mathcal{H}_{\widetilde{X}}.$$

Therefore the adjoint $\delta^* c = c\delta_\sharp$, and

$$\delta^*\delta = \delta_\sharp \otimes_{\widetilde{X}} \delta_\sharp, \qquad\qquad \langle f, (\delta^*\delta)g\rangle_{\widetilde{X}} = (\delta f)(\delta g).$$

Combining these identities, for every $f, g \in \mathcal{H}_{\widetilde{X}}$,

$$\langle f, \mathcal{S}g\rangle_{\widetilde{X}} = \langle f, \big(\mathcal{K} + \delta^*\delta\big)g\rangle_{\widetilde{X}}.$$

Non-degeneracy of $\langle\cdot,\cdot\rangle_{\widetilde{X}}$ on $\mathcal{H}_{\widetilde{X}}$ therefore gives

$$\mathcal{S} = \mathcal{K} + \delta^*\delta.$$

Rank and trace. Finally, the cross-product $\delta^*\delta = \delta_\sharp \otimes_{\widetilde{X}} \delta_\sharp \succcurlyeq 0$ has image contained in $\operatorname{span}(\delta_\sharp)$, and its only possible nonzero eigenvalue is $\|\delta_\sharp\|^2_{\widetilde{X}}$. Hence,

$$\operatorname{rank}(\delta^*\delta) \le 1, \qquad \operatorname{tr}(\delta^*\delta) = \|\delta_\sharp\|^2_{\widetilde{X}}.$$

Moreover,

$$D_{\mathcal{B}}(p_X, p_{\widetilde{X}}) = \sup_{h\in\mathcal{B}} |\delta h| = \sup_{\|h\|_{\widetilde{X}}\le 1} |\langle h, \delta_\sharp\rangle_{\widetilde{X}}| = \|\delta_\sharp\|_{\widetilde{X}}.$$

by Cauchy–Schwarz, attained at $h = \delta_\sharp / \|\delta_\sharp\|_{\widetilde{X}}$. Therefore,

$$\operatorname{tr}(\delta^*\delta) = D^2_{\mathcal{B}}(p_X, p_{\widetilde{X}}).$$

□

H.10 PROOF OF LEM. 2—CHARACTERIZING THE IDENTIFIED SET[1]

Lemma 2 (characterizing the identified set)**.** *Under Asm. 2 and 1a or 1c, $\mathcal{H}_{\text{PI}}$ is the covariance ball*

$$\mathcal{H}_{\text{PI}} = \left\{ h \in \mathcal{H}_X : \left\| h - h_{\text{erm}} \right\|_X^2 \leq \sigma^2 \gamma \right\},$$

within $\mathcal{H}_X$, centered at h_{erm} with radius $\sigma\sqrt{\gamma}$, and an effective confounding budget *of $\gamma := \frac{\gamma_{\boldsymbol{x}} \gamma_y}{1-\gamma_{\boldsymbol{x}}}$.*

Proof. With $\mathcal{B}$ the ball in the statement, the proof has three parts. (i) **Anchor:** all $q \in \mathcal{Q}_{\text{PI}}$ share the center h_{erm} and radius σ. (ii) **Necessity** ($\mathcal{H}_{\text{PI}} \subseteq \mathcal{B}$)**:** $h_\star^q \in \mathcal{B}$ for every q by bounding on budgets $\gamma_{\boldsymbol{x}}, \gamma_y$; (iii) **Sufficiency** ($\mathcal{H}_{\text{PI}} \supseteq \mathcal{B}$)**:** Any $h \in \mathcal{B}$ is achieved by some Gaussian $q^h \in \mathcal{Q}_{\text{PI}}$.

Anchor. Denote $\mathcal{V} := \operatorname{span}(\mathcal{H})$. The IPM test class $\mathcal{F} = \left\{ f^2 : f \in \mathcal{V} + \mathbb{R}Y \right\}$ in Asm. 2 matches p_{XY} and q_{XY} upto second moments of $(v(X), Y)$ for all $v \in \mathcal{V}$. Hence, all q induce a common risk

$$R_{\text{erm}}(h) = \mathbb{E}[Y^2] - 2\,\mathbb{E}[h(X)Y] + \mathbb{E}[h(X)^2], \qquad h \in \mathcal{H},$$

with common minimizer h_{erm} and minimum σ^2. Moreover,

$$\mathbb{E}_p[v(X) \cdot (Y - h_{\text{erm}}(X))] = 0 \qquad \forall v \in \mathcal{V}. \tag{♣}$$

Under Asm. 1a, this holds because $h_{\text{erm}} = \mathbb{E}_p[Y \,|\, X]$; under Asm. 1c, $\mathcal{V} = \mathcal{H}$ and $h_{\text{erm}} = \Pi_{\mathcal{H}} Y$.

For $v = 1 \in \mathcal{V}$, Eq. (♣) gives $\mathbb{E}[h_{\text{erm}}(X)] = \mathbb{E}[Y]$, confirming that $h_{\text{erm}} \in \mathcal{H}_X$ under mean-zero Y.

Necessity ($\mathcal{H}_{\text{PI}} \subseteq \mathcal{B}$)**.** Write Eq. (1) under some fixed $q \in \mathcal{Q}_{\text{PI}}$

$$Y = h_\star^q(X) + U_q + \xi_q, \qquad\qquad \xi_q \perp\!\!\!\perp (U_q, X, Z),$$

omitting the sub-/superscript q where unambiguous. Define

$$b^q := h_{\text{erm}} - h_\star^q \in \mathcal{H} - \mathcal{H} \subseteq \mathcal{V}, \qquad\qquad r_q := \mathbb{E}_q[Y \,|\, X] - h_{\text{erm}} \perp_{L_2(X)} \mathcal{V},$$

where $r_q \perp_{L_2(X)} \mathcal{V}$ holds because for every $v \in \mathcal{V}$, tower rule, moment matching, and Eq. (♣) give

$$\begin{aligned} \mathbb{E}_q[v(X) r_q(X)] &= \mathbb{E}_q[v(X)Y] - \mathbb{E}_q[v(X) h_{\text{erm}}(X)] \\ &= \mathbb{E}_p[v(X)Y] - \mathbb{E}_p[v(X) h_{\text{erm}}(X)] = 0. \end{aligned}$$

Conditioning the structural on X now yields,

$$\mathbb{E}_q[U \,|\, X] = \mathbb{E}_q[Y \,|\, X] - h_\star^q = \underbrace{r_q}_{\perp \mathcal{V}} + \underbrace{b^q}_{\in \mathcal{V}}. \tag{♠}$$

with r_q, b^q both X-centered. Now define:

$$a := \mathbb{E}_q[\operatorname{Var}_q(U \,|\, X)], \qquad\qquad c := \operatorname{Var}_q\big(\mathbb{E}_q[U \,|\, X]\big).$$

To show that $h_\star^q \in \mathcal{B}$, we need to show $\|b^q\|_X^2 \leq \sigma^2 \gamma$. This constitutes a chain of three inequalities,

$$\|b\|_X^2 \overset{(**)}{\leq} c \overset{(††)}{\leq} \frac{\gamma_{\boldsymbol{x}}}{1-\gamma_{\boldsymbol{x}}} a \overset{(‡‡)}{\leq} \frac{\gamma_{\boldsymbol{x}} \gamma_y}{1-\gamma_{\boldsymbol{x}}} \sigma^2 \leq \sigma^2 \gamma.$$

Here, (∗∗) holds by Pythagoras on Eq. (♠). And (††) follows from the X-budget

$$r^2_{U \sim X} := 1 - \frac{a}{\sigma_U^2} \leq \gamma_{\boldsymbol{x}} \quad \Longrightarrow \quad \sigma_U^2 \leq \frac{a}{1-\gamma_{\boldsymbol{x}}} \quad \Longrightarrow \quad c \overset{\text{total-var.}}{=} \sigma_U^2 - a \leq \frac{\gamma_{\boldsymbol{x}}}{1-\gamma_{\boldsymbol{x}}} a.$$

[1] By convention, set $\gamma := +\infty$ when $\gamma_{\boldsymbol{x}} = 1$ in Asm. 2, and $r^2 := 0$ whenever its denominator vanishes.

To establish (‡‡), we need to upper-bound a. The Y-budget gives,

$$r^2_{U\sim Y|X} \coloneqq 1 - \mathbb{E}[\mathrm{Var}(U\,|\,X,Y)]/\,a \;\le\; \gamma_y, \qquad\qquad \Longrightarrow \qquad\qquad (1-\gamma_y)\,a \;\le\; \mathbb{E}[\mathrm{Var}(U\,|\,X,Y)]. \qquad (\diamondsuit)$$

Now, condition on $X = \boldsymbol{x}$ and write $s(\boldsymbol{x}) \coloneqq \mathrm{Var}(U\,|\,X=\boldsymbol{x})$, so that $\mathbb{E}[s(X)] = a$. Note that $\mathbb{E}[\mathrm{Var}(U\,|\,Y,X)\,|\,X=\boldsymbol{x}]$ is the MMSE of predicting U from Y, so a *linear* MMSE upper-bounds it:

$$\mathbb{E}[\mathrm{Var}(U\,|\,Y,X)\,|\,X=\boldsymbol{x}] \le \frac{s(\boldsymbol{x})\,\sigma_\xi^2}{s(\boldsymbol{x})+\sigma_\xi^2}.$$

The RHS is concave in $s(\boldsymbol{x})$. For $a > 0$, Jensen and Eq. ($\diamondsuit$) give (conclusion immediate for $a = 0$),

$$(1-\gamma_y)\,a \le \frac{a\,\sigma_\xi^2}{a+\sigma_\xi^2}, \qquad\qquad \Longrightarrow \qquad\qquad a \le \gamma_y\,(a+\sigma_\xi^2).$$

This, combined with the below inequality gives $a \le \gamma_y\,\sigma^2$, finally establishing $\|b\|_X^2 \le \sigma^2\gamma$ via (‡‡).

$$a+\sigma_\xi^2 \overset{\text{total-var.}}{=} \mathbb{E}[\mathrm{Var}(Y\,|\,X)] \overset{\mathbb{E}[Y|X]\text{ optimality}}{\le} \min_{h\in\mathcal{H}} \mathbb{E}\Big[(Y-h(X))^2\Big] \quad =: \quad \sigma^2.$$

Sufficiency ($\mathcal{H}_{\mathrm{PI}} \supseteq \mathcal{B}$). Fix $h \in \mathcal{B}$, then define

$$b^h \coloneqq h_{\mathrm{erm}} - h, \qquad e^2 \coloneqq \big\|b^h\big\|_X^2 \le \sigma^2\gamma.$$

Suppose first that $\gamma_{\boldsymbol{x}} > 0$ and $e > 0$. Define,

$$\sigma_W^2 \coloneqq \frac{1-\gamma_{\boldsymbol{x}}}{\gamma_{\boldsymbol{x}}}e^2, \qquad \sigma_\xi^2 \coloneqq \sigma^2 - \sigma_W^2,$$

Since $e^2 \le \sigma^2\gamma$,

$$\Longrightarrow\; \sigma_W^2 \le \frac{1-\gamma_{\boldsymbol{x}}}{\gamma_{\boldsymbol{x}}}\sigma^2\gamma = \gamma_y\sigma^2 \le \sigma^2, \qquad (\heartsuit)$$

so $\sigma_\xi^2 \ge 0$. Sample independently

$$(X,Z) \sim p_{XZ}, \qquad W_h \sim \mathcal{N}(0,\sigma_W^2), \qquad \xi_h \sim \mathcal{N}(0,\sigma_\xi^2),$$

and define

$$U_h \coloneqq b^h(X) + W_h, \qquad Y = h(X) + U_h + \xi_h = h_{\mathrm{erm}}(X) + W_h + \xi_h.$$

This satisfies the structural equation with $h_\star^{q^h} = h$; U_h and ξ_h are centered, and $\xi_h \perp\!\!\!\perp (U_h, X, Z)$.

It remains to verify $q^h \in \mathcal{Q}_{\mathrm{PI}}$.

(a) Moment matching. For every $v \in \mathcal{V}$, $q^h_X = p_X$ and $Y - h_{\mathrm{erm}}(X) = W_h + \xi_h$ is centered, independent of X, and has variance σ^2. Thus

$$\mathbb{E}_{q^h}[v(X)Y] = \mathbb{E}_p[v(X)h_{\mathrm{erm}}(X)] = \mathbb{E}_p[v(X)Y],$$

by Eq. (♣), while

$$\mathbb{E}_{q^h}[v(X)^2] = \mathbb{E}_p[v(X)^2], \qquad \mathbb{E}_{q^h}[Y^2] = \mathbb{E}_p[h_{\mathrm{erm}}(X)^2] + \sigma^2 = \mathbb{E}_p[Y^2],$$

the latter by Pythagoras under p. First moments match similarly. Hence $D_{\mathcal{F}}(p_{XY}, q^h_{XY}) = 0$.

(b) The $\gamma_{\boldsymbol{x}}$ budget: Since $W_h \perp\!\!\!\perp X$,

$$a = \sigma_W^2, \qquad \sigma_U^2 = e^2 + \sigma_W^2 = e^2/\,\gamma_{\boldsymbol{x}},$$

and therefore

$$r^2_{U\sim X} = 1 - \frac{\sigma_W^2}{e^2/\,\gamma_{\boldsymbol{x}}} = 1 - (1-\gamma_{\boldsymbol{x}}) = \gamma_{\boldsymbol{x}} \le \gamma_{\boldsymbol{x}}.$$

(c) The γ_y budget: Conditional on X, observing Y is equivalent to observing $W_h + \xi_h$. Therefore

$$\mathrm{Var}_q(U \mid X = \boldsymbol{x}, Y) = \mathrm{Var}(W \mid W + \xi) = \frac{\sigma_W^2 \cdot \sigma_\xi^2}{\sigma_W^2 + \sigma_\xi^2} = \frac{\sigma_W^2\,(\sigma^2 - \sigma_W^2)}{\sigma^2},$$

whence

$$r^2_{U \sim Y \mid X} = 1 - \frac{\sigma_W^2\,(\sigma^2 - \sigma_W^2)/\sigma^2}{\sigma_W^2} = \frac{\sigma_W^2}{\sigma^2} \le \gamma_y,$$

using $\sigma_W^2 \le \gamma_y \sigma^2$ from Eq. (♡). Hence $q^h \in \mathcal{Q}_{\mathrm{pi}}$.

Edge cases. If $e = 0$ (equivalently $h = h_{\mathrm{erm}}$ in $\mathcal{H}_X$; this covers $\gamma_{\boldsymbol{x}} = 0$ or $\gamma_{\boldsymbol{y}} = 0$, where $\gamma = 0$ and the ball degenerates to $\{h_{\mathrm{erm}}\}$), take $U_h := 0$ and $\xi_h \sim \mathcal{N}(0, \sigma^2)$. Moment matching *(a)* holds as before, and both budgets *(b)–(c)* hold with $r^2 = 0$.[2]

Hence every $h \in \mathcal{B}$ is attained. □

Corollary 3 (characterizing the identified interval)**.** *Under Asm. 1c and Asm. 2, for $\boldsymbol{x} \in \mathrm{supp}(p_X)$,*

$$\mathcal{H}_{\mathrm{pi}}(\boldsymbol{x}) = \left\{h_{\mathrm{erm}}(\boldsymbol{x})\right\} \pm \sigma\sqrt{\gamma}\,\|g_{\boldsymbol{x}}\|_{L_2(X)},$$

where $g_{\boldsymbol{x}} \in \mathcal{H}_X$ is the $L_2(X)$-representer of $h \mapsto h(\boldsymbol{x})$, well defined by the non-degenerate $\langle \cdot, \cdot \rangle_X$.

Proof. For any $h \in \mathcal{H}_{\mathrm{pi}}$ as given by Lem. 2, evaluation at query point $\boldsymbol{x} \in \mathrm{supp}(p_X)$ follows

$$\begin{aligned}
|h(\boldsymbol{x}) - h_{\mathrm{erm}}(\boldsymbol{x})| &= \left|\langle h - h_{\mathrm{erm}},\, g_{\boldsymbol{x}} \rangle_{L_2(X)}\right|, && \text{(Reproducing identity.)} \\
&\le \|h - h_{\mathrm{erm}}\|_{L_2(X)} \cdot \|g_{\boldsymbol{x}}\|_{L_2(X)}, && \text{(Cauchy–Schwarz.)} \\
&\le \sigma\sqrt{\gamma}\,\|g_{\boldsymbol{x}}\|_{L_2(X)}, && \text{(From Lem. 2.)}
\end{aligned}$$

with equality at $h - h_{\mathrm{erm}} = \pm\sigma\sqrt{\gamma}\, g_{\boldsymbol{x}} / \|g_{\boldsymbol{x}}\|_{L_2(X)} \in \mathcal{H}$, membership by linearity of $\mathcal{H}$ (Asm. 1c). These two extremal functions are attained since they lie in the ball of Lem. 2 by construction.

Finally, since $\mathcal{H}_{\mathrm{pi}}$ is convex (a ball within linear $\mathcal{H}$ under Asm. 1c) and evaluation $h \mapsto h(\boldsymbol{x})$ is linear, the image $\mathcal{H}_{\mathrm{pi}}(\boldsymbol{x})$ is an interval, and therefore coincides with the full interval above. □

[2] Under the convention $r^2 := 0$ whenever its denominator vanishes (Asm. 2).

H.11 PROOF OF LEM. 3—AVERAGE ERROR AS A WORST-CASE SURROGATE

Lemma 3 (avg. error as worst-case surrogate)**.** *Let $\mathcal{H}(\boldsymbol{x}) \subset \mathbb{R}$ be a bounded interval at query $\boldsymbol{x}$. Under a squared-loss $\ell(a,b) \coloneqq (a-b)^2$, the following sandwich is tight in multiplicative constant 4:*

$$\overline{E}_{\boldsymbol{x}}(\mathcal{H}(\boldsymbol{x})) \;\le\; E^{+}_{\boldsymbol{x}}(\mathcal{H}(\boldsymbol{x})) \;\le\; 4 \cdot \overline{E}_{\boldsymbol{x}}(\mathcal{H}(\boldsymbol{x})).$$

Furthermore, under Asm. 1c and 2 the following inequality holds for the k-dim. ball $\mathcal{H}$ from Lem. 2:

$$\overline{E}(\mathcal{H}) \;\le\; E^{+}(\mathcal{H}) \;\le\; (k+3) \cdot \overline{E}(\mathcal{H}).$$

Proof. The lower bound is trivial (max. dominates mean). For upper bounds: (i) write closed forms of both interval metrics in center offset and half-width, (ii) compute a maximizer for their ratio, and (iii) derive a common bound for k-dim. ellipsoids.

Closed forms. Write $\mathcal{H}(\boldsymbol{x}) = [c-w, c+w]$ with interval center c and half-width $w \ge 0$. Let $d \coloneqq |c - h_\star(\boldsymbol{x})|$ be the center's offset from the truth. Under sq.-loss, the pointwise excess risk of a candidate y is $E_{\boldsymbol{x}}(y) = (y - h_\star(\boldsymbol{x}))^2$, so worst-case is attained at the endpoint farther from $h_\star(\boldsymbol{x})$,

$$E^{+}_{\boldsymbol{x}}(\mathcal{H}(\boldsymbol{x})) = \max_{|u|\le w}(d+u)^2 = (d+w)^2,$$

while a uniform draw decomposes into offset plus uniform second moment,

$$\overline{E}_{\boldsymbol{x}}(\mathcal{H}(\boldsymbol{x})) = \mathbb{E}_{u\sim\mathrm{Unif}[-w,w]}[(d+u)^2] = d^2 + \frac{1}{3}w^2. \qquad (\mathbb{E}[u]=0 \text{ kills cross term.})$$

Upper bound. Take $w > 0$, as otherwise the metrics are equal. Write their ratio in $t \coloneqq d/w \ge 0$,

$$\frac{E^{+}_{\boldsymbol{x}}(\mathcal{H}(\boldsymbol{x}))}{\overline{E}_{\boldsymbol{x}}(\mathcal{H}(\boldsymbol{x}))} = \frac{(t+1)^2}{t^2+1/3} =: f(t), \quad \Longrightarrow \quad f'(t) = \frac{2(t+1)(1/3-t)}{(t^2+1/3)^2},$$

so f increases on $t \in [0, 1/3]$ and decreases thereafter, with maximum

$$f(1/3) = \frac{(4/3)^2}{(1/3)^2+1/3} = \frac{16/9}{4/9} = 4.$$

Hence, $E^{+}_{\boldsymbol{x}}(\mathcal{H}(\boldsymbol{x})) \le 4 \cdot \overline{E}_{\boldsymbol{x}}(\mathcal{H}(\boldsymbol{x}))$, with equality exactly at the offset $d = w/3$.

k-dim. ellipsoids. Write an $L_2(X)$-ellipsoid $\mathcal{H} = c + \mathcal{A}\mathcal{B}^k_2$, with center c and k-dim. ball $\mathcal{B}^k_2$. Let

$$d \coloneqq \|c - h_\star\|_{L^2(X)}, \qquad v \coloneqq \frac{\mathrm{tr}(\mathcal{A}^*\mathcal{A})}{k+2}, \qquad R \coloneqq \sup_{\|u\|_2\le 1} \|Au\|_{L^2(X)}.$$

with v the average sq.-radius of a centered $L_2(X)$-ellipsoid. Hence, for $U \sim \mathrm{Unif}(\mathcal{B}^k_2)$,

$$\overline{E}(\mathcal{H}) = \mathbb{E}\Big[\|c + AU - h_\star\|^2_{L^2(X)}\Big] = d^2 + v.$$

Therefore,

$$\begin{aligned}
E^{+}(\mathcal{H}) &\le (d+R)^2, \\
&= d^2 + R^2 + 2dR, \\
&\le d^2 + R^2 + \frac{d^2R^2}{v} + v, && \text{(Cauchy-Schwarz } 2ab \le a^2+b^2 \text{ on } a = \tfrac{dR}{\sqrt{v}}, b = \sqrt{v}.) \\
&= \left(1 + \frac{R^2}{v}\right)(d^2+v), \\
&\le (1 + (k+2))(d^2+v), && \text{(Pythagoras } R^2 \le \mathrm{tr}(\mathcal{A}^*\mathcal{A}) = (k+2)v.) \\
&= (k+3) \cdot \overline{E}(\mathcal{H}).
\end{aligned}$$

The bound is sharp: let one semi-axis be 1, and remaining $k-1$ semi-axes tend to 0. Place $h_\star$ on the longest-axis line at distance v/R from the center. The ratio $E^{+}(\mathcal{H}) \Big/ \overline{E}(\mathcal{H})$ then tends to $k+3$. □

H.12 PROOF OF LEM. 4—REYNOLDS PROJECTION WITH DA

Lemma 4 (Reynolds projection with DA). *Under Asms. 1a–1b, and 4, post-DA ERM $h_{\widetilde{\text{erm}}}$ satisfies*

$$h_{\widetilde{\text{erm}}} = \mathcal{T}h_{\text{erm}} \in \mathcal{H}_{\text{inv}},$$

where $\mathcal{T}$ is the $L_2(X)$-orthogonal proj. onto the ambient inv. subspace $\mathcal{I}$ with $\mathcal{H}_{\text{inv}} = \mathcal{H} \cap \mathcal{I}$ (cf. §E). Furthermore, when the target $h_\star$ is $\mathcal{T}$-invariant, the post-DA bias $\tilde{b} \coloneqq h_{\widetilde{\text{erm}}} - h_\star$ similarly follows

$$\tilde{b} = \mathcal{T}b.$$

Proof. We structure the proof in two parts: (i) the conditional expectation over augmentation acts as a proj. onto $\mathcal{I}$; (ii) both the post-DA ERM and bias naturally follow this conditional expectation.

Reynolds projection with DA. Take the operator $(\mathcal{R}h)(\cdot) \coloneqq \mathbb{E}[h(X) \mid \widetilde{X} = \cdot]$ on $h \in L_2^0(X)$, defined $p_{\widetilde{X}}$-a.e., hence p_X-a.e. (Asm. 1b). $\mathcal{R}$ coincides with projection $\mathcal{T}$ under Asm. 4, $\mathcal{R}h = \mathcal{T}h$. To prove this, we need to show that it is invariant $\mathcal{R}h \in \mathcal{I}$, and its residual $h - \mathcal{R}h$ is orthogonal to $\mathcal{I}$.

(a) Invariance: Since group $\mathcal{T}$ is compact, the Haar measure p_T is left-invariant, i.e. for any $\boldsymbol{\tau} \in \mathcal{T}$,

$$\begin{aligned}
& \boldsymbol{\tau}T \overset{d}{=} T, && \text{(Def. of left-inv. } p_T\text{, follows from Asm. 4.)} \\
\Rightarrow\ & (\boldsymbol{\tau}T, X) \overset{d}{=} (T, X), && \text{(Follows from } T \perp\!\!\!\perp X\text{.)} \\
\Rightarrow\ & (\boldsymbol{\tau}\widetilde{X}, h(X)) \overset{d}{=} (\widetilde{X}, h(X)), && \text{(Apply } (\boldsymbol{\tau}, \boldsymbol{x}) \mapsto (\alpha(\boldsymbol{x}, \boldsymbol{\tau}), h(\boldsymbol{x})) \text{ on both sides.)} \\
\Rightarrow\ & \mathbb{E}[h(X) \mid \boldsymbol{\tau}\widetilde{X}] = \mathbb{E}[h(X) \mid \widetilde{X}], && \text{(Conditional expectation on both sides.)} \\
\Rightarrow\ & (\mathcal{R}h)(\boldsymbol{\tau}^{-1}\widetilde{X}) = (\mathcal{R}h)(\widetilde{X}). && \text{(Re-parameterize to } \mathcal{R}h\text{. } \mathcal{T} \text{ group} \Rightarrow \boldsymbol{\tau}^{-1} \text{ exists.)}
\end{aligned}$$

Because $\boldsymbol{\tau}$ is arbitrary, it follows that $(\mathcal{R}h)(X) = (\mathcal{R}h)(\widetilde{X})$, where the left-hand evaluation under p_X is again licensed by Asm. 1b. Also, $\mathbb{E}[(\mathcal{R}h)(\widetilde{X})] = \mathbb{E}[h(X)] = 0$, hence $\mathcal{R}h \in \mathcal{I}$. Immediately, if

$$\mathcal{R}h \in \mathcal{H} \quad \Longrightarrow \quad \mathcal{R}h \in \mathcal{H}_{\text{inv}}.$$

(b) Orthogonality: The residual $h - \mathcal{R}h$ is $L_2(X)$-orthogonal to every invariant test function $g \in \mathcal{I}$:

$$\begin{aligned}
\langle \mathcal{R}h, g \rangle_{L_2(X)} \coloneqq \mathbb{E}[(\mathcal{R}h)(X) \cdot g(X)] &= \mathbb{E}[(\mathcal{R}h)(\widetilde{X}) \cdot g(\widetilde{X})], && \text{(Both } \mathcal{R}h \text{ and } g \text{ are } \mathcal{T}\text{-inv.)} \\
&= \mathbb{E}[\mathbb{E}[h(X) \,|\, \widetilde{X}] \cdot g(\widetilde{X})], && \text{(Def. of } \mathcal{R}\text{.)} \\
&= \mathbb{E}[\mathbb{E}[h(X) \cdot g(\widetilde{X}) \,|\, \widetilde{X}]], && (g(\widetilde{X}) \text{ is } \widetilde{X}\text{-measurable.)} \\
&= \mathbb{E}[h(X) \cdot g(\widetilde{X})], && \text{(Tower property.)} \\
&= \mathbb{E}[h(X) \cdot g(X)] = \langle h, g \rangle_{L_2(X)}. && (\mathcal{T}\text{-inv. of } g.)
\end{aligned}$$

Subtracting the two sides gives $\langle h - \mathcal{R}h, g \rangle_{L_2(X)} = 0$ for all $g \in \mathcal{I}$. Together with part *(a)*, $\mathcal{R}h \in \mathcal{I}$ and $h - \mathcal{R}h \perp_{L_2(X)} \mathcal{I}$, therefore for all $h \in L_2^0(X)$,

$$\mathcal{R}h = \mathcal{T}h.$$

Application to ERM and Bias. Under Asm. 1a, $h_{\text{erm}}(X) = \mathbb{E}[Y \,|\, X]$ and $h_{\widetilde{\text{erm}}}(\widetilde{X}) = \mathbb{E}[Y \,|\, \widetilde{X}]$. Since Y is centered, all of $h_\star, h_{\text{erm}}, h_{\widetilde{\text{erm}}}$ (hence $b, \tilde{b}$) are centered. Further, $T \perp\!\!\!\perp (X, Y, U, \xi)$ by construction, so $Y - X - \widetilde{X}$ is a Markov chain. Via the tower rule,

$$\begin{aligned}
h_{\widetilde{\text{erm}}}(\widetilde{X}) = \mathbb{E}[Y \,|\, \widetilde{X}] &= \mathbb{E}\Big[\mathbb{E}[Y \,|\, X, \widetilde{X}] \,\Big|\, \widetilde{X}\Big], \\
&= \mathbb{E}\Big[\mathbb{E}[Y \,|\, X] \,\Big|\, \widetilde{X}\Big] = \mathbb{E}[h_{\text{erm}}(X) \,|\, \widetilde{X}] = (\mathcal{R}h_{\text{erm}})(\widetilde{X}).
\end{aligned}$$

Note that $h_{\widetilde{\text{erm}}}$ and $\mathcal{R}h_{\text{erm}}$ are $\mathcal{T}$-inv., hence they must also be identically equal in $L_2(X)$ (Asm. 1b), giving $h_{\widetilde{\text{erm}}} = \mathcal{R}h_{\text{erm}} = \mathcal{T}h_{\text{erm}}$ by part *(b)*. And since $h_{\widetilde{\text{erm}}} \in \mathcal{H}_X$ by Asm. 1a, part *(a)* places it in the class: $h_{\widetilde{\text{erm}}} \in \mathcal{H}_{\text{inv}}$. Finally, for a $\mathcal{T}$-inv $h_\star \in \mathcal{I}$, using $\mathcal{T}h_\star = h_\star$ and linearity of $\mathcal{T}$ on $L_2^0(X)$,

$$\mathcal{T}b \quad = \quad \mathcal{T}h_{\text{erm}} - \mathcal{T}h_\star \quad = \quad h_{\widetilde{\text{erm}}} - h_\star \quad = \quad \tilde{b}. \qquad \square$$

H.13 PROOF OF LEM. 5—AVERAGE REPRESENTER NORMS

Lemma 5 (avg. representer norms)**.** *Under Asms. 1b–1c, evaluation $h \mapsto h(\boldsymbol{x})$ admits the $L_2(X)$-representer $g_{\boldsymbol{x}} \in \mathcal{H}_X$ and the $L_2(\widetilde{X})$-representer $\tilde{g}_{\boldsymbol{x}} \in \mathcal{H}_{\widetilde{X}}$. Averaged over queries $X \sim p_X$ gives*

$$\mathbb{E}_X\Big[\,\|g_X\|^2_{L_2(X)}\Big] = k, \qquad\qquad \mathbb{E}_X\Big[\,\|\tilde{g}_X\|^2_{L_2(\widetilde{X})}\Big] = \operatorname{tr}(\mathcal{S}).$$

Equivalently, we can write the second moment operators of representers g_X and $\tilde{g}_X$ respectively as:

$$\mathbb{E}_X[\,g_X \otimes_X g_X\,] = \mathrm{Id}, \qquad\qquad \mathbb{E}_X[\,\tilde{g}_X \otimes_{\widetilde{X}} \tilde{g}_X\,] = \mathcal{S}.$$

Proof. Two steps give the result: (i) **expand** each representer in an orthonormal basis of its own centered class; (ii) **average** over $X \sim p_X$, reading each summand as a quadratic form of the shift operator.

Expansion. Fix an $L_2(\widetilde{X})$-orthonormal basis $\{\psi_i\}_{i=1}^k$ of the class $\mathcal{H}_{\widetilde{X}}$ (Asms. 1b–1c, cf. §H). Then,

$$\begin{aligned}
\tilde{g}_{\boldsymbol{x}} &= \sum_{i=1}^k \langle \tilde{g}_{\boldsymbol{x}}, \psi_i\rangle_{L_2(\widetilde{X})} \cdot \psi_i, && \text{(Parseval in } \langle\cdot,\cdot\rangle_{L_2(\widetilde{X})}.)\\
&= \sum_{i=1}^k \psi_i(\boldsymbol{x}) \cdot \psi_i, && \text{(Reproducing identity } \langle \psi_i, \tilde{g}_{\boldsymbol{x}}\rangle_{L_2(\widetilde{X})} = \psi_i(\boldsymbol{x}).)\\
\implies \quad \|\tilde{g}_{\boldsymbol{x}}\|^2_{L_2(\widetilde{X})} &= \sum_{i=1}^k \psi_i(\boldsymbol{x})^2. && \text{(Orthonormality of } \psi_i.)
\end{aligned}$$

Averaging. Take expectations over $X \sim p_X$ in each summand,

$$\mathbb{E}_X\Big[\,\psi_i(X)^2\Big] = \|\psi_i\|^2_{L_2(X)} = \langle \psi_i, \mathcal{S}\psi_i\rangle_{L_2(\widetilde{X})}, \qquad \text{(Def. of } \mathcal{S}.)$$

and sum over the basis to obtain a trace,

$$\mathbb{E}_X\Big[\,\|\tilde{g}_X\|^2_{L_2(\widetilde{X})}\Big] = \sum_{i=1}^k \langle \psi_i, \mathcal{S}\psi_i\rangle_{L_2(\widetilde{X})} = \operatorname{tr}(\mathcal{S}). \qquad \text{(Trace in any orthonormal basis.)}$$

Repeating the two steps in an $L_2(X)$-orthonormal basis $\{\psi_i\}_{i=1}^k$ of $\mathcal{H}_X$ replaces $\mathcal{S}$ by Id throughout. The averaged summand is then $\|\psi_i\|^2_{L_2(X)} = 1$, now resulting in $\mathbb{E}_X\Big[\,\|g_X\|^2_{L_2(X)}\Big] = \operatorname{tr}(\mathrm{Id}) = k$.

Second-moment version. For any $f, h \in \mathcal{H}_{\widetilde{X}}$,

$$\begin{aligned}
\langle f, \mathbb{E}_X[\,\tilde{g}_X \otimes_{\widetilde{X}} \tilde{g}_X\,]h\rangle_{L_2(\widetilde{X})} &= \mathbb{E}_X[\,f(X)h(X)\,], && \text{(Reproducing identity.)}\\
&= \langle f, \mathcal{S}h\rangle_{L_2(\widetilde{X})}, && \text{(Def. of } \mathcal{S}.)
\end{aligned}$$

so $\mathbb{E}_X[\,\tilde{g}_X \otimes_{\widetilde{X}} \tilde{g}_X\,] = \mathcal{S}$ on $\mathcal{H}_{\widetilde{X}}$, and identically $\mathbb{E}_X[\,g_X \otimes_X g_X\,] = \mathrm{Id}$ on $\mathcal{H}_X$. □